The Early Career and the Sources of Lifetime Inequality*

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Abstract

We study the years immediately following labor-market entry and their implications for measuring and understanding lifetime inequality. We estimate a life-cycle model with human-capital accumulation and frictional labor markets using data that follow workers from labor-market entry. We show that decompositions initialized several years after entry substantially overstate the role of initial conditions by reclassifying outcomes generated during the first years of work as initial heterogeneity. We also decompose the rise in early-career wage inequality and find an important role for sorting between human capital and match quality, as workers with higher human capital increasingly obtain better matches. This sorting arises because human capital raises job-finding rates, lowers separation risk, and shifts workers toward better offer distributions, generating incentives to work beyond those arising through wages alone.

Keywords: Life-Cycle Models, Labor Supply, Human Capital Accumulation

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1 Introduction

Standard life-cycle decompositions often use a common age in early adulthood as the base period, typically after most workers have completed their formal education (Huggett, Ventura, and Yaron, 2011; Ozkan, Song, and Karahan, 2023; Bick, Blandin, and Rogerson, 2024). For workers who enter the labor market directly after high school, however, ages such as 23 or 25 are already several years into their working lives. By then, these workers may have accumulated substantial experience, moved across jobs, experienced periods of nonemployment, and sorted into better or worse matches. Because these outcomes reflect labor-supply, human-capital accumulation, search, and mobility decisions, the years immediately following labor-market entry are important for measuring and understanding the sources of lifetime inequality.

This concern is substantive because, in the data, the early career is a period when hours worked rise, employment attachment increases, wages grow, and workers move across jobs. Interpreting these patterns requires going beyond either a standard life-cycle labor-supply model or a standard frictional-search model alone. Human-capital accumulation makes current work more valuable because it raises future productivity (Heckman, 1976; Imai and Keane, 2004; Fan, Seshadri, and Taber, 2024). Search frictions make early mobility valuable because workers move out of nonemployment and into better matches (Topel and Ward, 1992; Neal, 1999). Therefore, modeling these forces jointly is important for understanding early-career labor-market dynamics.

This paper studies the school-to-work transition and quantifies its importance for measuring and understanding the sources of lifetime inequality. To do so, we estimate a life-cycle model with an endogenous workweek, learning-by-doing, match quality, and frictional labor markets. The model is built around two state variables that summarize workers' labor-market position: human capital and match quality. Workers accumulate human capital through learning-by-doing, with hours worked entering the human-capital production function. Match quality evolves through random search in a frictional labor market, consistent with models in which early-career wage growth reflects job mobility and improved matches (Bagger, Fontaine, Postel-Vinay, and Robin, 2014; Menzio, Telyukova, and Visschers, 2016). Human capital affects not only the wage paid in the current job, as in standard life-cycle models of human-capital accumulation and labor supply (Heckman, 1976; Imai and Keane, 2004), but also job-finding rates, separation rates, and the mean of the offer distribution. The model also includes borrowing constraints and persistent heterogeneity in the disutility of work; the latter helps discipline hours dispersion, a key feature of life-cycle labor supply emphasized by Bick et al. (2024).

Measuring the early career in the data is central to our exercise. We estimate the model by indirect inference using data from the National Longitudinal Survey of Youth 1979, which we use to construct profiles with period zero at labor-market entry. We identify entry using the high-school graduation date and restrict attention to workers whose highest completed degree is a high-school diploma and who do not return to postsecondary education. Our profiles are indexed by quarters since labor-market entry rather than by age. With this construction, period zero in the model and in the data corresponds to the same economic moment: the start of working life. Each profile bin contains workers with the same potential experience rather than the same age, avoiding the composition problem that arises when workers enter the labor market at different ages.

Our first result is that the timing of measurement substantially changes the interpretation of lifetime inequality. Initial assets, human capital, match quality, and preference type explain 26.4 percent of lifetime earnings inequality when measured at labor-market entry, but 45.6 percent when measured five years later. In other words, within the same model, moving the base period from entry to year 5 raises the explained share by 19.2 percentage points. The year-5 base period is informative because it corresponds to age 23 for the average worker in our sample, making it comparable to the base period in Huggett et al. (2011).[1] State variables observed several years after entry already embed early-career shocks, labor-supply choices, human-capital accumulation, and match sorting. Starting later, therefore, reclassifies some of the inequality produced during the school-to-work transition as initial heterogeneity.

To understand why the early career is important, we decompose the growth in mean wages and wage inequality into different sources. Our second result is that early-career wage inequality grows primarily through sorting between human capital and match quality. We find that the covariance between human capital and match quality moves from negative at entry to positive after five years and accounts for most of the increase in wage variance. Neither skill dispersion nor match dis-

[1]Huggett et al. (2011) use age 23 as the base period and report that initial conditions account for roughly 60 percent of lifetime earnings inequality. We use our year-5 base period as the closest point of comparison, since it corresponds to age 23 for the average worker in our sample. However, this comparison is not one-for-one: differences relative to their estimate may reflect model structure, data sources (PSID versus NLSY), sample design (their broader set of workers and cohorts versus our focus on high-school graduates from specific cohorts), and the timing of measurement. Within our framework, the 19.2 percentage point gap between the year-5 and entry-based estimates isolates the effect of starting measurement after labor-market entry.

persion alone explains the rise in wage inequality. By contrast, mean wage growth reflects both skill accumulation and movement toward better matches: over the first 20 years, human capital contributes 0.326 log points and match quality contributes 0.322 log points.[2]

Lastly, we examine why identifying the mechanisms that shape early-career outcomes matters. In a model in which human capital affects wages alone, policies influence dynamic incentives to work through their effects on current or future pay (Imai and Keane, 2004; Keane and Rogerson, 2012). In our model, human capital also affects future employment and job opportunities, expanding the channels through which dynamic incentives arise.

Our third result is that human-capital accumulation is the dominant source of early-career labor-supply incentives, but only about half of this incentive operates through wages. We measure these incentives using the ratio of the marginal rate of substitution to the wage, as in (Imai and Keane, 2004; Keane and Rogerson, 2015). Values above one indicate that workers supply hours beyond the static optimum because current work generates future returns. A counterfactual economy with search frictions but no human-capital accumulation generates essentially no dynamic wedge, while a human-capital economy in which human capital affects wages alone generates only about half of the baseline wedge. The remaining incentive arises through human capital's effects on employment transitions and future job opportunities. A model that omits these channels would attribute the full dynamic return to work to future wages and could therefore mispredict labor-supply responses to interventions that affect hiring and separation frictions or search efficiency. Understanding the sources of the dynamic wedge is especially important during the early career, when the wedge is largest.

Together, our results show that the school-to-work transition is not merely a short prelude to the life cycle. It is a period in which labor supply, job mobility, human-capital accumulation, and match sorting jointly shape workers' trajectories and contribute to the formation of lifetime inequality. Our findings also show that frameworks initialized after this transition risk misclassifying endogenous early-career outcomes as predetermined heterogeneity.

[2]Like the lifetime-inequality decomposition above, the wage-dispersion decomposition is sensitive to the base period. Subsection 5.2 reports the decomposition from entry and from five years after entry; starting later shifts the apparent driver of wage-inequality growth from sorting toward human-capital dispersion.

Related Literature This paper contributes to literatures on lifetime inequality, early-career labor-market dynamics, and life-cycle labor supply. First, it relates to quantitative work on the sources of lifetime inequality (Huggett et al., 2011; Griffy, 2021; Ozkan et al., 2023; Gimpelson, 2024; Hosseini, Kopecky, and Zhao, 2026). Existing decompositions typically initialize workers at a common chronological age in early adulthood. We show that this timing matters because state variables observed several years after labor-market entry already embed early-career shocks, labor-supply choices, human-capital accumulation, and match sorting. Starting the decomposition years after labor-market entry reclassifies part of the inequality produced during the early career as initial heterogeneity. By aligning the model and data at labor-market entry, we distinguish inequality present at entry from inequality generated during the first years of work and show how the choice of base period affects the measured sources of lifetime inequality.

The closest paper on early-career inequality is Kaplan (2012), who shows that unemployment is central for understanding the decline in hours inequality early in the life cycle. We share the focus on young workers, hours dispersion, and early-career labor-market frictions. Kaplan studies hours inequality in an annual incomplete-markets model in which unemployment acts as an idiosyncratic labor-supply wedge. We study wages, hours, and employment in a monthly incomplete-markets model that explicitly accounts for employment, unemployment, and job-to-job transitions. Our framework allows hours to influence human-capital accumulation and labor-market prospects, which connects early-career hours dynamics to the formation of wage and lifetime earnings inequality.

Second, the paper relates to work on early-career wage growth, job mobility, and match quality. Young workers move across jobs and occupations as they search for better matches (Topel and Ward, 1992; Neal, 1999), and quantitative search models decompose early-career wage growth into experience, tenure, and match components (Bagger et al., 2014; Menzio et al., 2016; Gorry, 2016). In these models, experience typically increases deterministically with time and shapes wages, transitions, or learning about match quality. We make the rate of human-capital accumulation endogenous: hours worked raise future human capital, and human capital affects wages, separation risk, job-finding rates, and the quality of future offers. At the same time, we abstract from learning about match quality or worker ability.

Third, the paper relates to life-cycle models of labor supply with human-capital accumulation (Heckman, 1976; Weiss, 1986; Imai and Keane, 2004; Wallenius, 2011; Fan et al., 2024). In standard models, the dynamic return to current work operates primarily through the effect of human capital on future wages. In our model, this return also includes a labor-market-opportunity component: greater human capital raises job-finding rates, lowers separation risk, and improves the distribution of future job offers. Distinguishing these channels matters because they imply different labor-supply responses to policies that affect wages, employment stability, and job-search opportunities.

Michelacci and Pijoan-Mas (2012) also study intertemporal labor supply in a frictional labor market with human-capital accumulation. In their model, human capital raises job-arrival rates through a ranking mechanism: more productive workers are more likely to receive offers because applicants are ranked by productivity. We differ from their model in three relevant dimensions. First, human capital affects labor-market transitions directly, rather than only through a relative-ranking technology. Second, human capital shifts the mean of the match-quality distribution, thereby affecting not only the probability of receiving an offer but also the quality of future opportunities. Third, our model features a finite life cycle, endogenous hours, and multiple sources of heterogeneity, allowing us to quantify how these channels shape early-career labor supply and the formation of lifetime inequality.

Fourth, the paper relates to work on the school-to-work transition and youth labor markets. This literature studies how schooling, labor-market entry, and occupational choice are jointly determined (Keane and Wolpin, 1997; Lochner and Monge-Naranjo, 2011; Flinn and Mullins, 2015). We take completed schooling as given and study the evolution of labor-market outcomes after entry. Our focus on young workers also connects to work showing that they play a central role in aggregate labor-market dynamics (Shimer, 2001; Jaimovich, Pruitt, and Siu, 2013) and to Gorry (2013), who studies youth unemployment in a search model with experience accumulation.

The paper also relates to work emphasizing extensive-margin labor supply, fixed costs, and preference heterogeneity over the life cycle. Erosa, Fuster, and Kambourov (2016) show that fixed costs and preference heterogeneity are important for accounting for life-cycle labor supply, while Bick et al. (2024) show that preference heterogeneity is an important source of lifetime hours inequality. Our model incorporates heterogeneity in labor disutility and learning ability, as in Bick et al. (2024), but embeds these forces in a framework with job-switching costs and explicit employment, unemployment, and job-to-job transitions.

Roadmap

Section 2 describes the life-cycle model with human capital accumulation and job search frictions. Section 3 describes the data and our measurement choices, including labor market entry and the construction of quarters-since-graduation profiles. Section 4 describes our estimation strategy. We also use a statistical decomposition of total hours growth into its extensive and intensive components. Section 5 reports the quantitative results. Section 6 concludes.

2 Model

This section presents a life-cycle model used to separate inequality present at labor-market entry from inequality produced during the early career. Workers enter the model at the start of working life. They differ in assets, human capital, employment status, match quality if employed, labor disutility, and learning ability. After entry, their outcomes evolve through endogenous hours choices, learning by doing, employment and nonemployment spells, job-to-job mobility, and shocks to labor disutility.

The model has two central labor-market state variables: human capital and match quality. Human capital affects labor-market outcomes through three channels. First, it raises wages. Second, it changes transition rates by increasing job-offer arrival rates and lowering separation risk. Third, it shifts the distribution of match offers toward better matches. These channels allow current work to affect not only future wages, but also future labor-market prospects. Match quality matters because workers search across jobs and gradually sort into better matches.

We specify flexible functional forms for wages, transition rates, and offer distributions, and estimate the parameters governing the role of human capital and match quality for these outcomes. For example, as described below, wages are a power function in human capital and match quality, with estimated elasticities ρ_{ws} and ρ_{wm} that determine how each state variable translates into wages.

2.1 Setup

Time is discrete and measured at a monthly frequency. The monthly frequency allows the model to capture the job transitions, separations, and nonemployment spells that are especially common early in the career. Period $t = 0$ corresponds to labor-market entry.

Preferences Workers live for a finite horizon $t = 0, 1, \ldots, T$. In each period, they choose consumption c_t and, when employed, hours worked $h_t \in [0, 1]$. Lifetime utility is

$$\sum_{t=0}^{T} \beta^t \left(\ln(c_t) - \xi_t \frac{h_t^{1+\frac{1}{\gamma}}}{1+\frac{1}{\gamma}} \right), \qquad \beta \in (0, 1).$$

The parameter $\xi_t > 0$ governs the level of the disutility of work, while $\gamma > 0$ controls its curvature. We allow ξ_t to vary across workers and over the life cycle. This heterogeneity helps the model match dispersion in hours in mid-careers, as emphasized by Bick et al. (2024).

Labor Market At each date, a worker is either unemployed or employed in a match of quality m_t. Unemployed workers receive benefits $\bar{b}$ and search for jobs. Employed workers choose hours and search for new jobs on the job. The labor market is frictional and characterized by random search. All transition probabilities are logistic functions of human capital and, where relevant, match quality.[3]

An unemployed worker with human capital s_t receives a job offer with probability

$$\lambda^U(s_t) = \frac{1}{1 + \exp\left(\lambda_0^U - \rho_{\lambda s} \log s_t\right)}.$$

The parameter λ_0^U determines the baseline job-arrival rate, while $\rho_{\lambda s}$ governs how job finding varies with human capital. When $\rho_{\lambda s} > 0$, workers with more human capital receive job offers more frequently.

Offered match quality is drawn from a log-normal distribution. Let $F^U(m' \mid s_t)$ denote the distribution of match offers faced by unemployed workers. Then

$$F^U(m' \mid s_t) = \Phi\left(\frac{\log m' - \mu^U(s_t)}{\sigma^U}\right),$$

where

$$\mu^U(s_t) = \mu_0^U + \mu_1^U \frac{s_t^{1-\mu_2^U} - 1}{1 - \mu_2^U}.$$

The parameters $\mu^U(s_t)$ and σ^U are the mean and standard deviation of log offered

[3] We use logistic specifications for arrival and separation rates, since they keep transition probabilities in $[0, 1]$ for all state values. This ensures that continuation terms are well-defined expected values while allowing human capital and match quality to shift transition rates smoothly.

match quality, and $\Phi(\cdot)$ denotes the standard normal CDF. When $\mu_2^U = 1$, the Box-Cox term is defined as $\log s_t$. Human capital therefore shifts the mean of the offer distribution, so workers with more human capital draw better matches on average. If the worker rejects the offer, he remains unemployed. If he accepts the offer, he begins the next period employed in the accepted match. While unemployed, human capital depreciates at rate δ_u.

An employed worker is characterized by current human capital s_t and current match quality m_t. Labor income equals the wage per unit of labor times hours worked. The wage rate is

$$w(s_t, m_t) = w_0 \; s_t^{\rho_{ws}} \; m_t^{\rho_{wm}}.$$

The parameter w_0 determines the wage level, while ρ_{ws} and ρ_{wm} govern the elasticities of wages with respect to human capital and match quality.

Each period, an employed worker receives an outside offer with probability

$$\lambda^E(s_t, m_t) = \frac{1}{1 + \exp\left(\lambda_0^E - \rho_{\lambda s} \log s_t - \rho_{\lambda m} \log m_t\right)}.$$

The parameter λ_0^E determines the baseline offer-arrival rate, while $\rho_{\lambda s}$ and $\rho_{\lambda m}$ govern how outside-offer arrival varies with human capital and match quality.

Outside match offers are drawn from a log-normal distribution distinct from that faced by unemployed workers. Let $F^E(m' \mid s_t)$ denote the distribution of outside offers faced by employed workers. Then

$$F^E(m' \mid s_t) = \Phi\left(\frac{\log m' - \mu^E(s_t)}{\sigma^E}\right),$$

where

$$\mu^E(s_t) = \mu_0^E + \mu_1^E \; \frac{s_t^{1-\mu_2^E} - 1}{1 - \mu_2^E}.$$

The parameters $\mu^E(s_t)$ and σ^E are the mean and standard deviation of log offered match quality. Human capital shifts the mean of the outside-offer distribution, generating an offer-quality channel through which skill accumulation improves future job opportunities.

Employed workers also face exogenous separation risk. The probability of separation is

$$\pi(s_t, m_t) = \frac{1}{1 + \exp\Big(\pi_0 + \rho_{\pi s} \log s_t + \rho_{\pi m} \log m_t\Big)}.$$

The parameter π_0 determines the baseline separation rate. When $\rho_{\pi s} > 0$ and $\rho_{\pi m} > 0$, separation risk declines with human capital and match quality. Workers with more human capital and better matches are therefore less likely to lose their jobs. When separated, workers begin the next period unemployed.

Human Capital Employed workers accumulate human capital through learning by doing. Accumulation depends on current human capital s_t, match quality m_t, hours worked h_t, and learning efficiency α_t:

$$\mathcal{H}(s_t, m_t, h_t, \alpha_t) = \alpha_t s_t^{\rho_{ss}} m_t^{\rho_{sm}} h_t^{\rho_{sh}}.$$

The parameter $\alpha_t > 0$ determines the efficiency of learning by doing, while ρ_{ss}, ρ_{sm}, and ρ_{sh} are the elasticities of human capital accumulation with respect to current human capital, match quality, and hours worked. We allow workers to differ in learning efficiency, as described below.

Part of accumulated human capital is specific to the current job or occupation. Workers therefore lose human capital when they change jobs, either voluntarily or through separation.[4] We model this loss as a one-time adjustment cost that is convex in current human capital,

$$\phi(s_t) = B s_t^{\nu} \ .$$

with $B > 0$ and $\nu > 1$. The convexity implies that the loss rises more than proportionally with accumulated human capital.

This loss creates an interaction between job mobility and human capital accumulation. Current hours raise future human capital. Job changes and separations reduce the level of human capital carried into the next period, thereby lowering the continuation value of human-capital investment when future mobility is likely. This force is more important early in the career, when match quality is low and transition rates are high. As workers accumulate human capital and sort into better matches, mobility becomes more costly because more human capital is at risk,

[4] This assumption is consistent with evidence that specialized knowledge is imperfectly transferable across jobs and occupations. For example, Hernandez Martinez, Holter, and Pinheiro (2025) show that workers whose pre-displacement human capital is more specialized experience larger earnings losses after displacement.

generating endogenous lock-in and reducing job turnover with age.[5] In Appendix C, we assess the role of this assumption by simulating a counterfactual economy in which human capital is fully portable across jobs.

The law of motion for human capital is

$$s_{t+1} = \begin{cases} (1-\delta_e)s_t + \mathcal{H}(s_t, m_t, h_t, \alpha_t), & \text{if the worker remains in the same job,} \\ (1-\delta_e)s_t + \mathcal{H}(s_t, m_t, h_t, \alpha_t) - \phi(s_t), & \text{if the worker changes jobs or separates,} \end{cases}$$

where δ_e is the depreciation rate while employed. For unemployed workers, human capital depreciates at rate δ_u, with no new accumulation:

$$s_{t+1} = (1-\delta_u)s_t.$$

We restrict the human-capital state space to $s_t > 0$. In the quantitative implementation, we impose a small positive lower bound on human capital to ensure it remains positive.

Budget Constraint Both employed and unemployed workers can save and borrow using a one-period risk-free asset with net return r. Borrowing is subject to the constraint $a_{t+1} \geq \bar{a}_t$, where $\bar{a}_t$ is based on the natural debt limit implied by the present value of future income while unemployed and receiving unemployment benefits.[6] The per-period budget constraint is

$$c_t + a_{t+1} = \begin{cases} w(s_t, m_t)h_t + (1+r)a_t & \text{if employed,} \\ \bar{b} + (1+r)a_t & \text{if unemployed.} \end{cases}$$

Retirement Workers retire at age $T_R < T$. During retirement, they no longer supply labor and do not receive labor income. Instead, they receive a common retirement income Ret. The budget constraint then becomes

$$c_t + a_{t+1} = \mathrm{Ret} + (1+r)a_t, \qquad \text{for } t \geq T_R.$$

[5]This mobility-risk channel is related to Castex, Chow, and Dechter (2025), who show that anticipated automation risk reduces human-capital investment and slows wage growth. The source of risk differs. Their paper studies automation risk, whereas our model emphasizes mobility and separation risk in a frictional labor market.

[6]In the quantitative implementation, the borrowing constraint is $\bar{a}_t = \max\{a_t^{\text{nat}}, \bar{a}\}$, where a_t^{nat} is the natural debt limit and $\bar{a}$ is a calibrated lower bound.

Retirees cannot borrow against future retirement income, so $a_{t+1} \geq 0$ for $t \geq T_R$. The terminal condition is $a_{T+1} = 0$.

Initial Conditions and Type Heterogeneity At labor-market entry, workers draw initial assets, human capital, employment status, and labor-disutility type. Initial assets and human capital follow log-normal distributions. Initial employment status is drawn from a Bernoulli distribution with unemployment probability p_{u_0}.[7] Conditional on initial employment, workers also draw initial match quality from a log-normal distribution. Unemployed workers do not have an active match-quality state until they receive and accept a job offer.

Workers differ in their disutility of labor, ξ_t, and in the efficiency of human capital accumulation, α_t. The disutility parameter evolves over the life cycle according to an AR(1) process in logs:

$$\log \xi_{t+1} = (1 - \rho_\xi)\mu_\xi + \rho_\xi \log \xi_t + \varepsilon_{\xi,t+1}, \qquad \varepsilon_{\xi,t+1} \sim N(0, \sigma_\varepsilon^2),$$

where μ_ξ is the unconditional mean, $\rho_\xi \in (0,1)$ is the persistence parameter, and $\sigma_\varepsilon = \sigma_\xi \sqrt{1 - \rho_\xi^2}$ is the innovation standard deviation implied by the unconditional variance σ_ξ^2. The initial distribution of ξ_t is log-normal with mean μ_ξ and standard deviation $\sigma_{\xi,0} < \sigma_\xi$, so dispersion increases over the life cycle as workers receive idiosyncratic shocks. We interpret these shocks broadly as changes in willingness or ability to supply labor, including health-related shocks. See Hosseini et al. (2026) for an assessment of the importance of health inequality for lifetime earnings inequality.

Learning efficiency is linked to labor disutility according to

$$\log \alpha_t = C_0^\alpha + C_1^\alpha \log \xi_t,$$

or, equivalently, $\alpha_t = \exp(C_0^\alpha)\xi_t^{C_1^\alpha}$. Thus, learning efficiency is a deterministic function of the worker's current labor-disutility type and evolves with ξ_t. Because $\xi_t > 0$, this specification ensures that $\alpha_t > 0$.

The parameters C_0^α and C_1^α govern the relationship between learning efficiency and labor disutility. When $C_1^\alpha < 0$, workers with a higher disutility of work are also less efficient at accumulating human capital. This creates a reinforcing force:

[7] In the data, some workers are already employed at the time of high school graduation, which is the event we use to identify labor-market entry. The initial employment probability captures these jobs.

workers with high labor disutility supply fewer hours and accumulate less human capital through experience, and they also learn less efficiently while working. [8]

2.2 Workers' Problem and Value Functions

Workers make consumption, savings, and hours choices before labor-market shocks are realized. The timing within the period is as follows. Employed workers first choose hours and savings. They then face a separation shock. If not separated, they may receive an outside offer and decide whether to remain in their current job or accept the new match. Unemployed workers choose savings and may receive a job offer. Labor disutility for the next period is then realized. Human capital is updated using current hours, with an additional loss if the worker changes jobs or separates.

An employed worker at time t chooses consumption c_t, savings a_{t+1}, hours h_t, and, if an outside offer arrives, whether to remain in the current job or move to the new match. The value function depends on age, assets, human capital, match quality, and labor-disutility type ξ_t:

$$
\begin{aligned}
&V_t^W(a_t, s_t, m_t, \xi_t) = \\
&\quad \max_{c_t, a_{t+1}, h_t} \Bigg\{ u(c_t) - v(h_t, \xi_t) \\
&\quad + \beta\pi(s_t, m_t)\, \mathbb{E}_{\xi'} V_{t+1}^U(a_{t+1}, s_{t+1}^1, \xi') \\
&\quad + \beta\Big(1 - \pi(s_t, m_t)\Big)\lambda^E(s_t, m_t)\mathbb{E}_{m'}\Big[\max\Big\{\mathbb{E}_{\xi'} V_{t+1}^W(a_{t+1}, s_{t+1}^0, m_t, \xi'), \mathbb{E}_{\xi'} V_{t+1}^W(a_{t+1}, s_{t+1}^1, m', \xi')\Big\}\Big] \\
&\quad + \beta\Big(1 - \pi(s_t, m_t)\Big)\Big(1 - \lambda^E(s_t, m_t)\Big)\mathbb{E}_{\xi'} V_{t+1}^W(a_{t+1}, s_{t+1}^0, m_t, \xi')\Bigg\},
\end{aligned}
$$

subject to the budget constraint, the borrowing constraint, and $0 \leq h_t \leq 1$. The expectation over m' is taken with respect to $F^E(m' \mid s_t)$, and the expectation over ξ' is taken with respect to the transition process for labor disutility. Here s_{t+1}^0 and s_{t+1}^1 denote human capital next period for a worker who stays in the current job and for one who transitions (either by separation or by accepting an outside offer), respectively.

An employed worker accepts a new offer when the continuation value of the

[8] Huggett et al. (2011) and Bick et al. (2024) recover a positive correlation between labor disutility and learning efficiency. In their models, workers who dislike work more are more efficient at accumulating human capital. We recover a negative correlation.

new match, net of the human-capital loss from switching, exceeds the continuation value of remaining in the current match. The optimal hours choice reflects the within-period tradeoff between consumption and work effort, as well as the effect of current hours on future human capital. Human capital affects wages, transition rates, and offer quality. In Subsection 5.3, we decompose the quantitatively important components of labor-supply incentives.

An unemployed worker at time t chooses consumption and savings. If an offer arrives, the worker chooses whether to accept the offer or remain unemployed. The value function is

$$\begin{aligned}
&V_t^U(a_t, s_t, \xi_t) = \\
&\quad \max_{c_t, a_{t+1}} \Bigg\{ u(c_t) \\
&\quad + \beta\lambda^U(s_t)\mathbb{E}_{m'}\left[\max\left\{ \mathbb{E}_{\xi'} V_{t+1}^U(a_{t+1}, s_{t+1}^U, \xi'), \mathbb{E}_{\xi'} V_{t+1}^W(a_{t+1}, s_{t+1}^U, m', \xi') \right\} \right] \\
&\quad + \beta\Big(1 - \lambda^U(s_t)\Big)\mathbb{E}_{\xi'} V_{t+1}^U(a_{t+1}, s_{t+1}^U, \xi') \Bigg\},
\end{aligned}$$

subject to the unemployed budget constraint and the borrowing constraint. The expectation over m' is taken with respect to $F^U(m' \mid s_t)$. As usual, acceptance is characterized by a reservation match quality. Here s_{t+1}^U denotes human capital next period.

A retired worker at time $t \geq T_R$ chooses consumption and savings. The value function depends only on age and assets:

$$V_t^R(a_t) = \max_{c_t, a_{t+1}} \left\{ u(c_t) + \beta V_{t+1}^R(a_{t+1}) \right\},$$

subject to the retiree budget constraint and the no-borrowing constraint. At the retirement boundary, working-age continuation values are identified with the retirement value, $V_{T_R}^U(a, s, \xi) = V_{T_R}^W(a, s, m, \xi) = V_{T_R}^R(a)$. Thus, when $t = T_R - 1$, all continuation values in the employed and unemployed Bellman equations reduce to $V_{T_R}^R(a_{T_R})$, independently of employment status, human capital, match quality, and labor-disutility type. The terminal condition is $V_{T+1}^R(a_{T+1}) = 0$.

The model contains two sources of variation. The first is heterogeneity in initial conditions at labor-market entry: assets, human capital, employment status, match quality for initially employed workers, and labor disutility. The second is shocks that arrive over the life cycle: employment and job-offer shocks, match-quality draws, and shocks to labor disutility. The quantitative analysis uses the model to separate the contribution of these initial conditions from inequality generated after entry through labor supply, human-capital accumulation, mobility, and sorting.

The model keeps several features parsimonious in order to focus on the interaction between human-capital accumulation and search. Separations are exogenous, wages are given by a reduced-form wage function rather than by bargaining, and retirement income is modeled in reduced form. To capture lifecycle patterns driven by factors exogenous to the model core, we multiply labor disutility, offer arrival rates, separation rates, wages, and human capital accumulation by an age-scaling factor $D(\zeta, t)$, where

$$D(\zeta, t) = \begin{cases} (1+\zeta)^t & \text{if } \zeta < 0, \\ \log(1+\zeta\, t) + 1 & \text{if } \zeta \geq 0, \end{cases}$$

and t denotes age in months. These scalings allow the model to match empirical age profiles without attributing them to the endogenous mechanisms. Together, these choices preserve what is central to the paper: current hours raise future human capital, human capital affects both wages and labor-market prospects, and workers sort across jobs of different match quality over the early career.[9]

3 Measurement: Labor-Market Entry and Profiles

This section describes how we align model time with data time. The model begins at labor-market entry, so the empirical profiles used in estimation must also begin at labor-market entry rather than at a fixed chronological age. We therefore identify the date at which workers enter the labor market and construct profiles indexed by quarters since entry. These profiles are the empirical counterparts of the model life cycle.

We use data from the National Longitudinal Survey of Youth 1979 and 1997 (NLSY79 and NLSY97). These surveys are well-suited for this purpose because

[9] Blandin (2018) shows that allowing the growth in learning ability to vary with age brings lifecycle models with human capital accumulation through learning-by-doing closer to the data. He also shows that the learning-by-doing model is similar to the Ben-Porath model in explaining lifecycle profiles of earnings, wages, and hours. However, the two differ in their predictions for earnings growth dispersion. This is because, in the learning-by-doing model, workers cannot separate learning from earning as the return to human capital declines late in the career. Since our model focuses on the early career, the late-career distinction between learning-by-doing and Ben-Porath is second-order for our purposes. We adopt the same spirit of allowing parameters to vary with age.

they combine detailed education histories with long panel information on employment, hours, wages, and jobs. The NLSY79 follows a nationally representative sample of 12,686 individuals born between 1957 and 1964 who were aged 14-22 in 1979. The NLSY97 follows a nationally representative sample of 8,984 individuals born between 1980 and 1984 who were aged 12-17 in 1997.

3.1 Measuring Labor-Market Entry

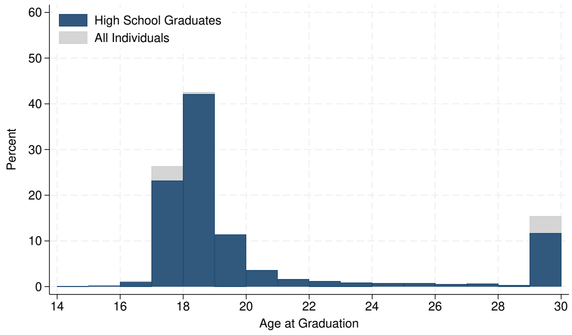
We measure labor-market entry using the high-school graduation date and restrict the sample to individuals whose highest completed degree is a high-school diploma and who do not return to formal education after graduation. For these workers, high-school graduation provides a clear empirical counterpart to labor-market entry.[10] This sample restriction allows period zero in the model and period zero in the data to refer to the same economic moment: the start of working life.

The NLSY79 and NLSY97 contain the information needed to construct this entry date. In addition to their long panel structure and detailed labor-market information, the surveys record educational attainment and the timing of degree receipt. Graduation dates are directly available in the NLSY97 but must be constructed in the NLSY79. For the NLSY79, we use two sets of questions. First, respondents report whether they received a diploma, which diploma they received, and the date it was awarded. Second, they report the highest degree attained and the date it was obtained. When conflicting dates are reported for the same diploma, we manually cross-check responses using additional information, including the highest degree attained up to that survey wave and the reported reason for leaving school.

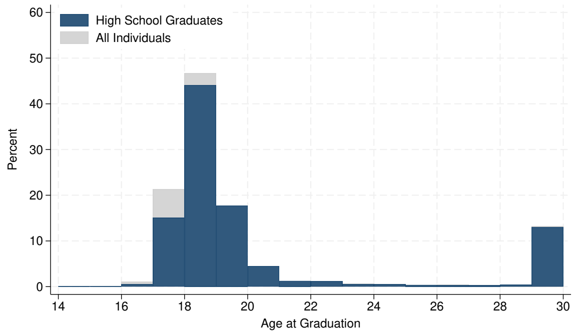
Figure 1, Panels (a) and (b), shows the distribution of labor-market entry in the NLSY79 and NLSY97. Each panel presents two histograms: the distribution for individuals whose highest educational attainment is a high-school diploma and, in shaded bars, the distribution for all respondents for whom a secondary-school completion date can be constructed, including those who subsequently complete additional schooling. These distributions are shown before imposing the additional panel-duration, military-service, and observation-window restrictions described below.

Two patterns stand out. First, the age at high-school graduation is dispersed.

[10] For individuals who did not complete high school, entry is harder to define because it may occur at dropout or later. For individuals with education beyond high school, entry may occur after high-school graduation, college graduation, or the completion of professional or postgraduate degrees.



(a) NLSY79



(b) NLSY97

Figure 1: High School Graduation Distribution

Note: The figure plots the age distribution at high school graduation for individuals in the NLSY79 and NLSY97 samples. The lighter bars represent all individuals, while the darker bars highlight those whose highest educational attainment is a high school diploma. Ages below 14 and above 30 are grouped. For the NLSY79, it is not possible to distinguish between GED certificates and high school diplomas; both are treated equivalently. For the NLSY97, when both GED and high school graduation dates are available, the earlier date is used.

In the NLSY79, the mean, median, and standard deviation of labor-market entry are 19.64, 18.34, and 4.40 years, respectively; in the NLSY97, they are 19.14, 18.52, and 2.39 years. Second, by ages 21 and 25, individuals in the NLSY79 have, on average, 2.13 and 5.51 years of post-entry labor-market experience, respectively; the corresponding numbers in the NLSY97 are 1.97 and 5.34 years.

These patterns imply that age-based profiles may suffer from two problems. First, they mix workers with different amounts of post-entry labor-market history. Workers of the same age may differ substantially in how long they have been employed, unemployed, accumulating human capital, and searching across jobs. A decomposition initialized at a fixed age may therefore condition on outcomes already produced after labor market entry. Second, when age-based profiles begin at young ages and average across workers, some individuals enter the sample only at later ages. This delayed entry changes the composition of the sample over the profile. Our approach avoids both problems by indexing workers by quarters since high-school graduation.

3.2 Data and Sample Restrictions

Having identified labor-market entry, we use the NLSY employment histories to construct quarters-since graduation profiles. Employment status, employer identifiers, and workweek are available at a weekly frequency. Other variables, including wages and job characteristics, are reported at the survey-wave level for each job. We assign these variables to the corresponding survey year and map them to calendar weeks within that year. We then aggregate the weekly data to a quarterly frequency.

After defining the target population and analysis frequency, we impose additional sample restrictions. Table 1 lists all restrictions and their impact on sample size. We sequentially apply the following criteria: (i) completion of the 12th grade; (ii) reporting a valid high-school graduation date; (iii) graduation between ages 16 and 24, to restrict the sample to a typical life-cycle trajectory; (iv) no military service; (v) at least 10 years of panel participation in the NLSY79 or 5 years in the NLSY97; and (vi) observations available for either the first or second year following graduation. After applying all restrictions, the final sample consists of 3,341 respondents from the original 12,686 individuals in the NLSY79 and 1,563 respondents from the original 8,984 individuals in the NLSY97.[11]

	NLSY79		NLSY97	
	Ind.	% Sample	Ind.	% Sample
Initial Sample	12,686	100.0%	8,984	100.0%
Final Sample	3,341	26.3%	1,563	17.4%
Restrictions applied				
Completed 12th grade	6,009	47.4%	2,386	26.6%
Reported a valid graduation date	5,521	43.5%	2,112	23.5%
Graduated between ages 16 and 24	5,072	40.0%	2,024	22.5%
Did not serve in the military	3,906	30.8%	1,911	21.3%
Met panel duration requirements	3,563	28.1%	1,803	20.1%
Observed 1st or 2nd year post-graduation	3,341	26.3%	1,563	17.4%

Table 1: Sample Selection Criteria and Final Sample Size

We impose an additional sample restriction when constructing the life-cycle profiles used as calibration targets. In particular, we focus on individuals with sustained labor-market attachment, defined as 520-5,200 annual hours. This restriction ensures that the empirical targets are aligned with the margins represented in the model. We do not model labor-force participation as a separate margin, and the nonemployment state includes both unemployment and some nonparticipation. Therefore, the hours' profiles used in the calibration should be interpreted as pro-

[11] Appendix A.1 reports descriptive statistics to evaluate potential selection concerns arising from our sample restrictions. The table compares three groups from the NLSY79: the full cohort, all high-school graduates, and the final analysis sample. As expected, high-school graduates differ substantially from the full cohort, but the final sample closely resembles the broader population of high-school graduates.

files for attached workers, rather than for all workers or the full population. Similar restrictions are used by Kaplan (2012) and Bick et al. (2024).[12] Section 4.5 examines how much changes in the composition of the profile sample influence the life-cycle profiles. These changes reflect both permanent exits from the panel and temporary movements into and out of eligibility due to the annual-hours restriction. Appendix A.3 reports the corresponding profiles without the annual-hours restriction.

Mean profiles. Let i index individuals and q index quarters since high-school graduation. We construct four mean profiles.

- *Employment rate*: The fraction of weeks in quarter q that individual i spends employed, averaged across individuals. This profile is unconditional and measured in levels. In the data, nonemployment includes unemployment and some nonparticipation, so the profile should be interpreted as an employment profile for moderately attached workers. The model counterpart is the fraction of the quarter spent employed.
- *Workweek*: Average weekly hours worked in quarter q, conditional on positive employment. We focus on hours in the main job. Hours are divided by 168, the weekly time endowment, so the measure lies between zero and one. The model counterpart is hours conditional on employment. We explored this measure with a broader measure that includes hours worked across all jobs, and the profiles look similar.
- *Wages*: The average real hourly wage across employed weeks in quarter q, conditional on employment. We construct real wages using the CPI. The mean wage profile is normalized by its value in the first post-entry quarter, so the targeted profile is wage growth relative to labor-market entry. The model counterpart is the average wage conditional on employment, normalized in the same way.
- *Cumulative jobs*: The number of distinct jobs held by individual i up to the end of quarter q. This profile is a stock measured at the end of the quarter rather than a flow averaged over the quarter. The model counterpart is the cumulative number of accepted jobs since labor-market entry.

[12] See Rogerson and Wallenius (2013) and Erosa et al. (2016) for approaches that explicitly model labor-force participation and extensive-margin labor supply.

Dispersion profiles. For each mean profile, we compute the corresponding cross-sectional standard deviation. The standard deviations of wages and workweek are computed conditional on employment and are measured in logs. The standard deviation of employment is computed from the individual-level fraction of weeks employed in the quarter and is measured in levels. The standard deviation of cumulative jobs is computed from the number of distinct jobs held by the end of the quarter and is also measured in levels. These dispersion profiles discipline cross-sectional heterogeneity in wages, hours, employment histories, and mobility.

Transition rates. We construct three transition rates from week-to-week changes in labor-market status: job-to-job, employment-to-nonemployment, and nonemployment-to-employment. For each transition type, we count transitions between consecutive weeks and divide by the number of individuals at risk in the origin state. A job-to-job transition is a direct change in job identifier between two consecutive employed weeks. A separation is a transition from employment to nonemployment. Job finding is a transition from nonemployment to employment.

Weekly hazard rates are converted to quarterly rates by compounding over thirteen weeks:

$$\lambda^{\text{quarter}} = 1 - \left(1 - \lambda^{\text{week}}\right)^{13}.$$

We then apply a four-quarter moving average to smooth the transition series. Because directly measured transition rates may be inconsistent with observed employment dynamics due to time-aggregation bias or measurement error, we adjust the separation rate following Shimer (2012).[13]

4 Estimation

This section describes our estimation procedure. The estimation chooses parameters so that the model reproduces the profiles constructed in Section 3. Because

[13] We use the measured quarterly employment profile and job-finding rate to construct an accounting-based adjusted separation rate. Specifically, we apply the employment accounting relationship $e_q = (1-\pi_q)e_{q-1} + f_q(1-e_{q-1})$, where e_q is the average fraction of the quarter spent employed, f_q is the job-finding rate, and π_q is the adjusted separation rate. Solving for π_q gives

$$\pi_q = \frac{e_{q-1} + f_q(1-e_{q-1}) - e_q}{e_{q-1}}.$$

Because e_q is a within-quarter employment measure rather than an end-of-quarter stock, this adjustment should be interpreted as an accounting-based consistency measure rather than an exact stock-flow inversion. We use this adjusted series in estimation instead of the directly measured separation rate.

these profiles start at labor-market entry, they discipline the distribution of state variables at entry. Because they also follow workers after entry, they also discipline the evolution of labor market outcomes over the early and mid-career.

Our estimation proceeds in two steps. First, we assign values to parameters that are directly observable, conventional in the literature, or difficult to identify separately. Second, we estimate the remaining parameters by simulated minimum distance, minimizing the distance between the empirical profiles and their model-implied counterparts. A similar estimation strategy is used by Fan et al. (2024).

To compute model moments, we solve and simulate the model at monthly frequency. We aggregate simulated histories to quarters and compute the same objects as in the data: mean profiles, dispersion profiles, and transition-rate profiles. The targeted mean profiles are employment, workweek, wages, and cumulative jobs. The targeted dispersion profiles are the cross-sectional standard deviations of employment, workweek, wages, and cumulative jobs. The transition-rate targets are job-finding rates, separation rates, and job-to-job transition rates. As in the data, wages and workweeks are measured conditional on employment, employment is measured as the fraction of the quarter spent employed, and cumulative jobs are measured as an end-of-quarter stock.

In the estimation, we use only profiles from a single cohort to ensure an internally consistent sample with parameters specific to that cohort. We use the NLSY79 because it provides the longest post-entry panel. The NLSY97 is used to assess the stability of the empirical profiles across cohorts. Appendix A.2 shows that the main life-cycle patterns are broadly similar across cohorts. We restrict the targeted profiles to the first 120 quarters after labor-market entry, corresponding to 30 years. This restriction keeps the estimation focused on the early and middle career, rather than on the transition to retirement.

4.1 Externally Calibrated Parameters

Table 2 reports the externally calibrated parameters. These parameters govern demographics, institutions, preferences, initial assets, and a small set of labor-market objects pinned down outside the simulated minimum-distance procedure.

The life cycle lasts $T = 840$ months (70 years). Workers enter the labor market at period zero, retire after $T_R = 540$ months (45 years), and spend the remaining 25 years in retirement. The interest rate is set to $r = 0.0025$ per month, corresponding to an annual rate of approximately 3 percent. The wage scale is normalized to

Parameter	Description	Value
Demographics and Institutional		
T	Total life periods (70 years, monthly)	840
T_R	Retirement period (45 years of work life)	540
r	Interest rate (monthly, 3% annual)	0.0025
w_0	Wage scale normalization	1.00
κ	Retirement income	0.18
$\bar{b}$	Unemployment benefit	0.12
Preferences		
β	Discount factor (monthly)	0.994
γ	Frisch elasticity of labor supply	0.55
Initial Asset Conditions		
p_{A_0}	Probability of zero initial assets	0.16
μ_{A_0}	Mean of log initial assets	1.26
σ_{A_0}	Std. dev. of log initial assets	0.64
$\bar{a}$	Borrowing limit	-4.54
Labor Market		
p_{e_0}	Initial employment rate	0.75
δ_e	Human capital depreciation (employed)	0.0025
$\rho_{\lambda m}$	Offer rate elasticity w.r.t. match quality	0.00

Table 2: Externally Calibrated Parameters

$w_0 = 1$.

Retirement income is set to the benefit level $\kappa = 0.18$. This value is constructed by applying a 60 percent replacement rate to a normalized wage of 1 and a workweek of 0.3, so that $\kappa = 0.6 \times 1 \times 0.3 = 0.18$. Unemployment benefits are set analogously to the benefit level $\bar{b} = 0.12$, using a 40 percent replacement rate, so that $\bar{b} = 0.4 \times 1 \times 0.3 = 0.12$.

Preferences are logarithmic in consumption. We set the monthly discount factor to $\beta = 0.994$ and the Frisch elasticity of labor supply to $\gamma = 0.55$, close to micro estimates around 0.5 for high-school-educated workers.[14]

Initial assets are calibrated from the 1983 Survey of Consumer Finances, restricting the sample to high-school-educated household heads aged 25 or younger. Net worth is expressed in 1993 dollars and scaled by average monthly entry earnings in the NLSY79. Because the model does not allow workers to begin with negative assets, we censor nonpositive net-worth observations at zero when constructing the initial asset distribution. The distribution is therefore a mixture of a point

[14]Other life-cycle models with endogenous labor supply and human capital accumulation estimate larger values. For example, Imai and Keane (2004) report $\gamma \approx 3.82$, while Wallenius (2011) finds $\gamma \approx 1.13$. Their models are at an annual frequency, while ours is at a monthly frequency.

mass at zero and a lognormal distribution for strictly positive asset holdings. The point mass, $p_{A_0} = 0.16$, equals the empirical share of households with nonpositive net worth. The lognormal parameters, $(\mu_{A_0}, \sigma_{A_0}) = (1.26, 0.64)$, match the mean and variance of log net worth among households with strictly positive net worth. We retain the uncensored net-worth distribution when calibrating the borrowing limit. The borrowing limit, $\bar{a} = -4.54$, is set to the first percentile of scaled net worth. Initial assets are assumed to be independent of initial human capital and match quality.

The initial employment rate p_{e_0} is set to match the average employment rate observed in the first quarter after high-school graduation. Human capital depreciates while employed at rate $\delta_e = 0.0025$ per month, approximately 3 percent annually. We set $\rho_{\lambda m} = 0$, so that offer arrival rates (both when unemployed and employed) depend on human capital but not directly on match quality, differing only in their level (λ^U and λ^E).

The initial levels of the job-finding, job-to-job, and separation rates are pinned down within the estimation routine. For each candidate parameter vector, we choose the intercepts of the transition functions so that the model matches the first-quarter job-finding rate, job-to-job transition rate, and separation rate observed in the NLSY79. This step is necessary because the initial transition rates depend jointly on the initial distributions of human capital and match quality, as well as on the elasticities that map these distributions to the transition rates.

4.2 Internally Estimated Parameters

Table 3 reports the internally estimated parameters. These parameters govern the initial distribution of human capital and match quality, transition rates, wages, human-capital accumulation, match-quality offers, preference heterogeneity, and residual age effects.

The parameter vector is estimated by simulated minimum distance. Let $\hat{m}$ denote the vector of empirical moments and $f(\theta)$ the corresponding vector of simulated model moments, computed from a synthetic panel of $N = 10{,}000$ individuals. The estimator solves

$$\hat{\theta} = \arg\min_{\theta} \Big[\hat{m} - f(\theta)\Big]' \Big[\hat{m} - f(\theta)\Big],$$

so the weighting matrix is the identity matrix.

The objective function is minimized using a multi-stage global search procedure inspired by Arnoud, Guvenen, and Kleineberg (2019). We first evaluate the objective at 2^{12} Sobol quasi-random points over the parameter space. We then select the 2^6 best-performing points and use each as a starting value for local derivative-free optimization. Finally, we form convex combinations of the best estimate and the remaining local minima and restart local optimization from those points. This final step probes the regions between competing local optima. The estimate $\hat{\theta}$ is the parameter vector with the lowest objective value across all stages.

Standard errors are computed using a bootstrap-based sandwich estimator. On the data side, we draw 10,000 bootstrap samples by resampling individuals with replacement and recomputing all empirical moments. The covariance matrix of these bootstrap moment vectors provides $\hat{\Omega}_{\text{data}}$, the estimated sampling variance of the empirical moments. On the model side, we approximate the Jacobian

$$\hat{J} = \frac{\partial f(\theta)}{\partial \theta'}$$

by central finite differences around $\hat{\theta}$, re-solving the model after perturbing each parameter. Because the estimated parameters differ by several orders of magnitude, we use a parameter-specific step size equal to one percent of each parameter's magnitude.

Since the weighting matrix is the identity, the estimated variance-covariance matrix is

$$\widehat{\text{Var}}(\hat{\theta}) = \left(\hat{J}'\hat{J}\right)^{-1} \hat{J}'\left(\hat{\Omega}_{\text{data}} + \hat{\Omega}_{\text{sim}}\right)\hat{J}\left(\hat{J}'\hat{J}\right)^{-1}.$$

The middle matrix collects two sources of variation. The first, $\hat{\Omega}_{\text{data}}$, is the sampling variance of the empirical moments. The second, $\hat{\Omega}_{\text{sim}}$, arises because model moments are computed from a finite simulated sample and therefore constitute an additional source of estimation error. We estimate $\hat{\Omega}_{\text{sim}}$ directly as the covariance of the simulated moment vector across 100 independent simulations at $\hat{\theta}$, each using N = 10,000 individuals and an independent set of random draws. We report the square roots of the diagonal elements as standard errors.[15]

[15] The table also reports λ^U, λ^E, and π_0, which are chosen inside the estimation routine to match first-quarter transition rates. Their standard errors are computed using the same procedure as for the other estimated parameters.

4.3 Identification

The targeted profiles jointly discipline four groups of model objects: the initial distribution at labor-market entry, human-capital accumulation, labor-market transition rates, and match-quality offers. The parameters governing these objects are estimated jointly, and there is no one-to-one mapping between individual moments and individual parameters. The identification discussion below describes which features of the data are most informative about each mechanism.

The initial wage, workweek, employment, and dispersion profiles discipline the distribution of state variables at entry. The parameters μ_{s_0}, σ_{s_0}, μ_{m_0}, σ_{m_0}, and $\rho_{sm,0}$ determine the initial joint distribution of human capital and match quality, while $\sigma_{\xi,0}$ determines the initial dispersion in labor-disutility types around μ_{ξ}. The estimated correlation between initial human capital and match quality is negative, $\rho_{sm,0} = -0.459$, so workers with higher human capital are not necessarily in better matches at labor-market entry. Together, initial preference heterogeneity and the negative correlation between human capital and match quality help the model match the substantial dispersion in workweeks at entry. The negative correlation is also central to the sorting results below: the model begins with negative sorting between human capital and match quality and then generates positive sorting during the early career.

Wage growth and workweek growth discipline human-capital accumulation. The parameters ρ_{ss}, ρ_{sm}, and ρ_{sh} govern how future human capital depends on current human capital, match quality, and hours worked. The estimated elasticity with respect to hours is $\rho_{sh} = 0.430$. The unemployment depreciation rate, $\delta_u = 0.009$, exceeds the employed depreciation rate, so nonemployment spells reduce human capital faster.

The transition-rate profiles discipline the transition-rate channel. The elasticities $\rho_{\lambda s}$, $\rho_{\pi s}$, and $\rho_{\pi m}$ determine how human capital and match quality affect job finding, job-to-job mobility, and separations. Human capital modestly raises offer arrival rates, with $\rho_{\lambda s} = 0.101$, and strongly lowers separation risk, with $\rho_{\pi s} = 2.167$. Match quality also lowers separation risk, with $\rho_{\pi m} = 0.872$. These estimates show that the transition-rate channels through which human capital improves labor-market prospects are quantitatively active in the estimated parameter set.

The wage and mobility profiles also discipline the offer-quality channel. The parameters governing the match-quality offer distributions determine whether workers with higher human capital draw from better distributions of future matches.

For employed workers, the mean of the match-quality offer distribution rises strongly with human capital, with $\mu_1^E = 0.915$. For unemployed workers, the corresponding slope is positive but smaller, $\mu_1^U = 0.108$. These parameters allow human capital to affect not only the probability of receiving offers but also the quality of those offers

The wage elasticities ρ_{ws} and ρ_{wm} determine how human capital and match quality translate into wages. Both state variables are important for wages, with $\rho_{ws} = 0.322$ and $\rho_{wm} = 0.674$. Match quality plays a central role in wage levels, while human capital affects wages both directly and indirectly through transition rates and offer quality.

The job-to-job transition profile and the number of jobs held over the career discipline the parameters governing losses at job transitions. The loss function $\phi(s) = Bs^{\nu}$ determines how much human capital a worker forfeits when changing jobs. The estimated curvature ν implies convex losses, meaning that workers with more accumulated experience lose proportionally more from a transition, generating endogenous lock-in that reduces job mobility as the career progresses. This mechanism helps the model reproduce the declining job-to-job transition rates observed in the data without relying solely on the direct effect of match quality on separations.

Parameter	Description	Value	Std. Error
Initial Distributions			
μ_{s_0}	Mean of log initial human capital	0.250	0.012
$\log \sigma_{s_0}$	Log std. dev. of initial human capital	-0.699	0.021
μ_{m_0}	Mean of log initial match quality	0.424	0.029
$\log \sigma_{m_0}$	Log std. dev. of initial match quality	-0.389	0.043
$\rho_{sm,0}$	Correlation of initial $\log s_0$ and $\log m_0$	-0.459	0.039
Labor Market Parameters			
λ^U	Offer arrival intercept (unemployed)	0.638	0.017
λ^E	Offer arrival intercept (employed)	1.932	0.047
π_0	Separation intercept	2.178	0.035
$\rho_{\lambda s}$	Offer rate elasticity w.r.t. human capital	0.101	0.006
$\rho_{\pi s}$	Separation rate elasticity w.r.t. human capital	2.167	0.045
$\rho_{\pi m}$	Separation rate elasticity w.r.t. match quality	0.872	0.030
ρ_{ws}	Wage elasticity w.r.t. human capital	0.322	0.008
ρ_{wm}	Wage elasticity w.r.t. match quality	0.674	0.020
Human Capital Parameters			
ρ_{ss}	HC accumulation elasticity w.r.t. human capital	0.039	0.002
ρ_{sm}	HC accumulation elasticity w.r.t. match quality	0.019	0.003
ρ_{sh}	HC accumulation elasticity w.r.t. hours	0.430	0.011
δ_u	HC depreciation rate (unemployed)	0.009	< 0.001
B	Scale parameter in $\phi(s)$	0.008	< 0.001
ν	Curvature parameter in $\phi(s)$	4.405	0.045
Match Quality Distributions			
μ_0^E	Intercept in $\mu^E(s)$	-0.032	0.003
μ_1^E	Scale parameter in $\mu^E(s)$	0.915	0.031
μ_2^E	Box-Cox parameter in $\mu^E(s)$	1.194	0.047
$\log \sigma^E$	Log std. dev. of outside offer quality	-1.692	0.084
μ_0^U	Intercept in $\mu^U(s)$	0.183	0.007
μ_1^U	Scale parameter in $\mu^U(s)$	0.108	0.009
μ_2^U	Box-Cox parameter in $\mu^U(s)$	1.147	0.070
$\log \sigma^U$	Log std. dev. of unemployed offer quality	-2.258	0.075
Type Heterogeneity			
$\mathbb{E}\xi$	Mean labor disutility (levels)	78.270	1.424
$\log \sigma_\xi$	Log std. dev. of stationary $\log \xi$	-0.332	0.007
ρ_ξ	Persistence of $\log \xi$	0.992	0.001
$\log \sigma_{\xi,0}$	Log std. dev. of initial $\log \xi$	-1.159	0.039
C_0^α	Intercept relating $\log \alpha$ to $\log \xi$	1.783	0.037
C_1^α	Slope relating $\log \alpha$ to $\log \xi$	-1.242	0.006
Age Effects			
ζ_ξ	Age decay in labor disutility	-0.0009	0.0001
ζ_λ	Age decay in offer arrival rate	0.0017	0.0001
ζ_π	Age decay in separation rate	-0.0012	0.0001
ζ_w	Age decay in wages	-0.0004	0.0001
ζ_α	Age decay in HC accumulation	-0.0003	< 0.001

Table 3: Internally Estimated Parameters

The parameters governing preference heterogeneity are disciplined primarily by the level and dispersion of workweeks. We estimate and report the level mean $\mathbb{E}\xi$ directly in Table 3; the log-location parameter entering the AR(1) is then $\mu_\xi = \log(\mathbb{E}\xi) - \frac{1}{2}\sigma_\xi^2$. Labor disutility is persistent, with $\rho_\xi = 0.992$, and its dispersion helps the model reproduce cross-sectional differences in hours worked. The parameters C_0^α and C_1^α govern heterogeneity in human-capital accumulation ability. Together, these parameters allow workers to differ in both their willingness to work and their return to work.

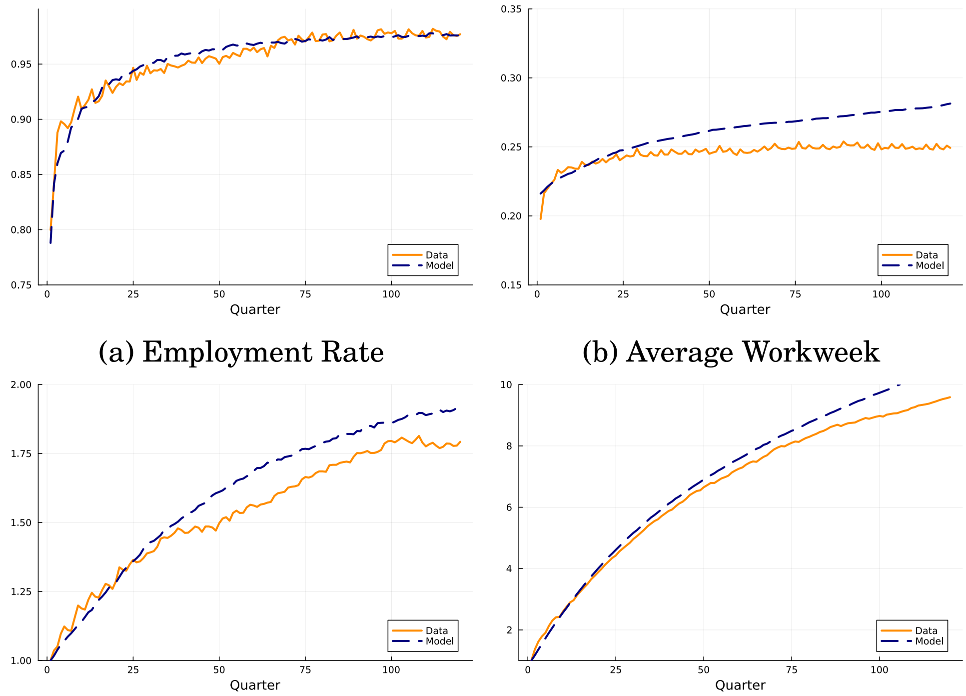
Finally, the age-effect parameters capture smooth residual variation in wages, transition rates, labor disutility, and human-capital accumulation that is not explained by human capital, match quality, or preference heterogeneity. These effects are small. They improve the fit of life-cycle profiles while leaving the main economic mechanisms to operate through human capital, match quality, and labor-market frictions. We also include log wage measurement error with variance 0.2, which helps match the level of wage dispersion; Bick et al. (2024) estimates a comparable measurement-error variance of 0.17.

4.4 Targeted Moments: Model Fit

We first assess the fit of the estimated model to the targeted profiles. The purpose of this exercise is to evaluate the model's overall fit and to show that the estimated structure reproduces the early-career patterns that discipline the main mechanisms. The model produces employment rate growth, workweek growth, wage growth, job mobility, cross-sectional dispersion, and transition rates.

Figure 2 compares the model-implied and empirical mean profiles. Panels 2a and 2b report the employment rate and the average workweek. In the data, both employment and the workweek rise during the first years after labor-market entry and then flatten. The model reproduces the rise and the subsequent plateau in both profiles, although it understates the steep initial increase in the workweek. This profile is important because workweek growth disciplines the human-capital accumulation channel.

Panel 2c reports the average wage profile. The data show sustained wage growth after labor-market entry, and the model closely tracks this upward profile. Panel 2d reports the cumulative number of jobs held. The data show rapid job accumulation early in the career, followed by a slower increase as workers gain labor-market experience. The model reproduces this pattern, capturing the high rate of early-career reallocation and the subsequent decline in mobility.



(a) Employment Rate

(b) Average Workweek

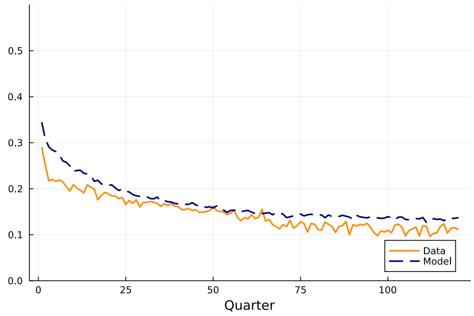
(c) Average Wage

(d) Number of Jobs

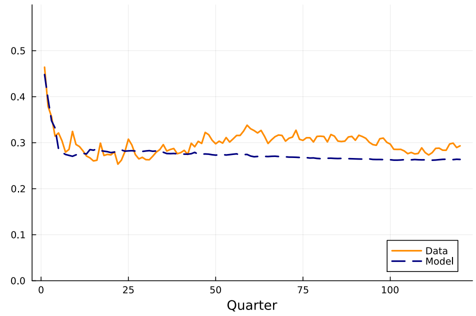
Figure 2: Comparison of Model-Implied and Data Profiles

Figure 3 compares the model-implied and empirical dispersion profiles. Panels 3a and 3b report the cross-sectional standard deviations of employment and log workweeks. Employment dispersion is high early in the career and then stabilizes. Workweek dispersion declines sharply during the first years after entry, a pattern emphasized by Kaplan (2012). The model reproduces the broad decline in workweek dispersion. This moment is important because it disciplines the initial correlation between human capital and match quality.

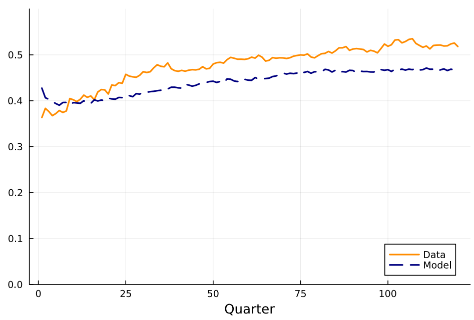
Panel 3c reports wage dispersion. In the data, wage dispersion rises over the early career, and the model generates the same upward pattern. Panel 3d reports dispersion in cumulative jobs. The data exhibit substantial heterogeneity in early-career mobility, with some workers changing jobs frequently and others settling into more stable employment relationships quickly. The model reproduces the increasing dispersion in cumulative job histories.



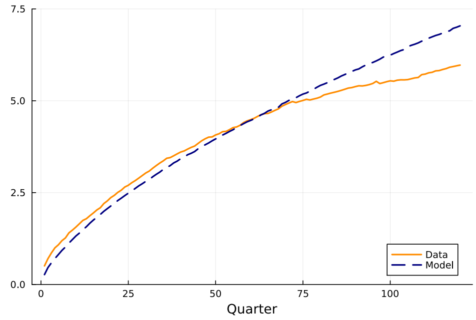
(a) Employment Rate (Std)



(b) Average Workweek (Std)



(c) Average Wage (Std)

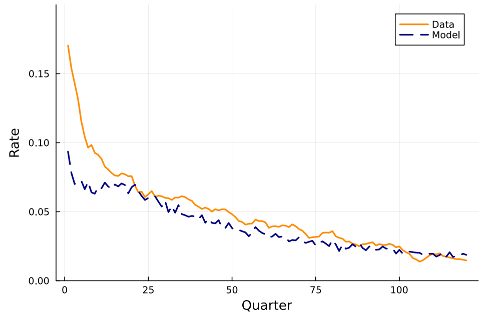


(d) Number of Jobs (Std)

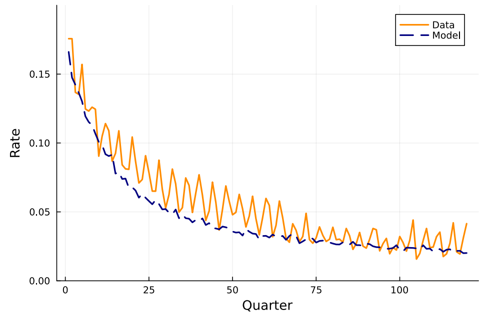
Figure 3: Comparison of Model-Implied and Data Profiles (Standard Deviations)

Figure 4 compares the model-implied and empirical transition-rate profiles. Panel 4a reports job-to-job transitions. The data show high job-to-job mobility at labor-market entry followed by a steady decline. The model captures the downward slope, although it understates the very high level of job-to-job transitions at entry. Panel 4b reports employment-to-nonemployment transitions. Separations decline with experience in both the model and the data. The adjusted separation series is less smooth because it is backed out from the employment accounting identity described in Section 3. Panel 4c reports nonemployment-to-employment transitions. The job-finding rate declines modestly over the career, both in the model and in the data.

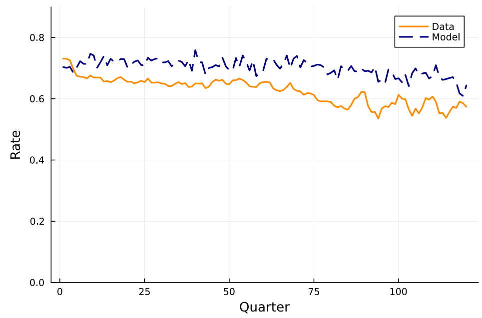
Overall, the model reproduces the main targeted patterns: rising employment, rising workweeks, sustained wage growth, declining job mobility, rising wage dispersion, declining workweek dispersion, and falling transition rates. The main limitations are that the model understates the steep initial rise in workweeks and the high level of job-to-job transitions immediately after labor-market entry. While these limitations should be kept in mind, we conclude that the model captures the key early-career dynamics needed for the lifetime-inequality decompositions below.



(a) Job to Job Rate



(b) Job to Unemployment Rate



(c) Unemployment to Job Rate

Figure 4: Comparison of Model-Implied and Data Profiles (Standard Deviations)

4.5 Untargeted Moment: Decomposing Hours into Intensive and Extensive Margins

We next evaluate the model using a decomposition of hours growth into intensive and extensive margins. This decomposition is treated as an untargeted moment. Total hours can rise because individuals work more weeks during a period or because they work longer weeks when employed. Many life-cycle models emphasize the extensive margin, focusing on employment or participation as the main source of hours dynamics. Our exercise, by contrast, indicates that most of the increase in hours after labor-market entry comes from longer workweeks rather than from additional weeks worked. This fact motivates the model's explicit treatment of the workweek as an intensive-margin choice.

Decomposition. Let H_t denote mean hours worked t quarters after labor-market entry. Let S_t be the set of individuals who are employed and have complete hours information in quarter t, and let $N_t = |S_t|$ be the number of individuals in S_t. Mean hours are

$$H_t = \frac{1}{N_t} \sum_{i \in S_t} H_{it} = \sum_{i \in S_t} \omega_{it} H_{it}, \qquad \omega_{it} = \frac{1}{N_t}. \tag{1}$$

Individual hours are the product of the average workweek and the number of weeks worked during the quarter,

$$H_{it} = h_{it} n_{it},$$

where h_{it} is average hours per week and $n_{it} > 0$ is the number of weeks worked in quarter t.[16]

To quantify the contribution of each margin, we decompose the change in mean hours between adjacent quarters. Let $S_t^C = S_t \cap S_{t-1}$ denote continuers, the individuals observed in both periods. Let $S_t^E = S_t \setminus S_{t-1}$ and $S_t^L = S_{t-1} \setminus S_t$ denote entrants and leavers. Then the change in mean hours can be written as

$$
\begin{aligned}
H_t - H_{t-1} = & \underbrace{\sum_{i \in S_t^C} \omega_{it} \left(h_{it} - h_{it-1}\right) n_{it-1}}_{\text{intensive margin}} \\
& + \underbrace{\sum_{i \in S_t^C} \omega_{it} h_{it-1} \left(n_{it} - n_{it-1}\right)}_{\text{extensive margin (1): within-continuers}} \\
& + \underbrace{\sum_{i \in S_t^E} \omega_{it} h_{it} n_{it} - \sum_{i \in S_t^L} \omega_{it-1} h_{it-1} n_{it-1} + \sum_{i \in S_t^C} \left(\omega_{it} - \omega_{it-1}\right) h_{it-1} n_{it-1}}_{\text{extensive margin (2): composition effect}} \\
& + \underbrace{\sum_{i \in S_t^C} \omega_{it} \left(h_{it} - h_{it-1}\right) \left(n_{it} - n_{it-1}\right)}_{\text{quadratic term}}.
\end{aligned}
\tag{2}
$$

The first term is the intensive margin: the change in total hours due to changes in the workweek, holding weeks worked fixed at their lagged value. The second term is the within-continuers extensive margin: the change in total hours due to changes in weeks worked, holding the workweek fixed at its lagged value. The third term is the composition margin, which captures entry into and exit from the employed sample, as well as changes in sample weights. The fourth term is the interaction between changes in the workweek and changes in weeks worked.[17]

We cumulate each component over quarters. Total hours growth between entry and quarter T is

$$H_T - H_0 = \sum_{t=1}^{T} (H_t - H_{t-1}). \tag{3}$$

[16] For individuals with positive weeks worked, we compute $h_{it} = H_{it}/n_{it}$.

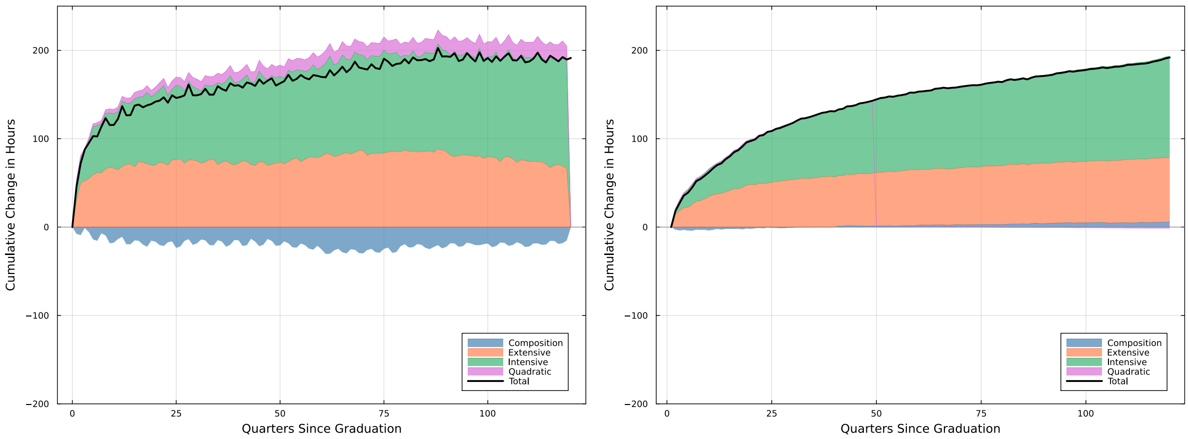
[17] Appendix B derives equation (2). The decomposition is a discrete-time Laspeyres decomposition: changes are evaluated relative to period $t-1$, and the quadratic term appears because changes in weeks worked and the workweek may occur simultaneously.

Each component of equation (2) can be summed over time and expressed either in levels or as a contribution to cumulative hours growth relative to the initial level. The share of cumulative hours growth is

$$\frac{\sum_{t=1}^{T} c_{kt}}{H_T - H_0}, \tag{4}$$

where c_{kt} denotes the period-t contribution of component k. The shares reported in Table 4 use this latter normalization.

Results. Figure 5 reports the decomposition in the data and in the model. The data show that the intensive margin accounts for most of the increase in hours after graduation. Over the first ten years, longer workweeks account for nearly 100 additional hours, while additional weeks worked account for about 60 hours. The composition and quadratic terms are small. The model reproduces the same qualitative pattern. Although it understates some of the total increase in hours, it captures the dominance of the intensive margin and the limited role of the composition and quadratic terms.[18]



(a) Data Decomposition (b) Model Decomposition

Figure 5: Intensive vs. Extensive Margin Decomposition

Table 4 reports the percentage contribution of each margin to cumulative hours growth at year-equivalent horizons of 5, 10, 15, and 20 years after graduation. The decomposition is computed at the quarterly frequency, and these horizons correspond to 20, 40, 60, and 80 quarters since labor-market entry. In the data, the intensive margin accounts for 57.0 percent of cumulative hours growth after five years, 57.3 percent after ten years, 59.6 percent after fifteen years, and 56.5 percent after twenty years. The extensive margin among continuers is also important,

[18]All results use the same sample as in Appendix A.2, pooling the NLSY79 and NLSY97 cohorts.

but changes in sample composition partially offset its contribution. The quadratic term is small.

The model closely reproduces the relative importance of the intensive and extensive margins, even though this decomposition is not targeted in estimation. After ten years, the intensive margin accounts for 56.5 percent of cumulative hours growth in the model, compared with 57.3 percent in the data. After twenty years, the corresponding shares are 58.2 percent in the model and 56.5 percent in the data. This untargeted fit shows that the model reproduces the empirical decomposition of early-career hours growth between the intensive and extensive margins: most of the increase comes from longer workweeks rather than from greater labor-market attachment alone. The counterfactual exercises below identify the mechanisms generating this pattern.

Table 4: Contributions of Intensive and Extensive Margins: Data vs. Model

Years	Intensive	Extensive (1)	Extensive (2)	Quadratic
Data				
5	57.0%	49.4%	-11.6%	5.2%
10	57.3%	45.0%	-9.4%	7.1%
15	59.6%	47.1%	-14.9%	8.2%
20	56.5%	45.9%	-10.2%	7.8%
Model				
5	51.8%	49.5%	-2.1%	0.8%
10	56.5%	43.1%	0.2%	0.3%
15	57.7%	41.0%	1.5%	-0.2%
20	58.2%	40.6%	1.7%	-0.5%

Note: Extensive (1) refers to within-continuers margin. Extensive (2) refers to composition margin. All values represent percentage contributions to cumulative growth up to year X.

5 Quantitative Experiments

In this section, we use the estimated model to study three related questions. First, how is lifetime inequality decomposed into initial conditions and subsequent shocks, and how does this decomposition depend on when measurement begins? Second, what drives the growth and dispersion of wages and earnings over the early career? Third, why do the mechanisms through which human capital operates matter for early-career labor-supply incentives? The three exercises produce a unified message: the early career is important for both the formation and measurement of lifetime inequality, and the mechanisms operating during this period shape workers' dynamic incentives to supply labor.

5.1 Lifetime Inequality: Initial Conditions vs Shocks

We now use the estimated model to study the sources of lifetime inequality. The exercise follows Huggett et al. (2011), who decompose lifetime dispersion into the share accounted for by initial conditions and the share generated by subsequent shocks. Our exercise differs in one important respect: our initial period corresponds to labor-market entry, rather than a fixed age cutoff.[19] This distinction is central. If the first years after entry are a period of rapid skill accumulation, job shopping, and sorting, then an age-based decomposition treats outcomes generated during the school-to-work transition as predetermined initial conditions.

Let Y_i denote a lifetime outcome for worker i, and let X_{i0} denote the vector of initial states. We use the law of total variance,

$$\mathrm{Var}(Y_i) = \mathrm{Var}\left(\mathbb{E}[Y_i \mid X_{i0}]\right) + \mathbb{E}\left[\mathrm{Var}(Y_i \mid X_{i0})\right],$$

and report the fraction

$$\frac{\mathrm{Var}\left(\mathbb{E}[Y_i \mid X_{i0}]\right)}{\mathrm{Var}(Y_i)}.$$

This statistic measures the share of lifetime inequality accounted for by differences already present at the base period; the residual is attributed to subsequent shocks.

We compute the decomposition for three lifetime outcomes: lifetime earnings (the discounted sum of earnings over the working life), lifetime wealth (lifetime earnings plus assets at the base period), and lifetime hours (total hours over the working life). We report three nested definitions of initial conditions. The first includes initial assets and human capital, (a_0, s_0). The second adds initial employment and match quality, m_0. The third adds the initial preference type, ξ_0.[20]

Table 5 reports the decomposition for two base periods: labor-market entry and five years after entry. In each case, the outcome is measured from the base period onward. Initial conditions at labor-market entry account for only a modest share of lifetime inequality. Under the broadest definition – assets, human capital,

[19] Huggett et al. (2011) use age 23.

[20] In the simulation, the conditional expectation $\mathbb{E}[Y_i \mid X_{i0}]$ is computed by grouping workers over the discretized state space. Continuous assets, human capital, and match quality are assigned to adjacent grid points using linear interpolation weights. Workers without an initial job are assigned to a separate no-match state. Preference types are already discrete.

match quality, and preference type – initial conditions account for 26.4 percent of lifetime earnings inequality, 33.8 percent of lifetime wealth inequality, and 17.7 percent of lifetime hours inequality.

Table 5: Sources of Lifetime Inequality

	From Entry			From Year 5
Statistic	Initial a_0, s_0	+ match m_0	+ prefs ξ_0	a_0, s_0, m_0, ξ_0
Lifetime earnings	0.032	0.145	0.264	0.456
Lifetime wealth	0.129	0.230	0.338	0.476
Lifetime hours	0.052	0.100	0.177	0.226

Note: The table reports decompositions computed in the estimated model for the simulated analog of the sample of high school graduates. The calibration profiles impose the annual-hours restriction described in Section 3, and the results should be interpreted as applying to attached workers. For each base period, lifetime outcomes are measured from that base period onward. Lifetime wealth is defined as assets held in the base period plus the present discounted value of labor earnings from that period onward.

The nested columns show that match quality and preference heterogeneity each contribute substantially. In particular, the preference type and the corresponding hours results speak to Bick et al. (2024): preference heterogeneity is the most important addition to the initial state for explaining lifetime hours inequality, raising the explained share from 10.0 to 17.7 percent. Even with this addition, the overall contribution of initial heterogeneity to lifetime hours inequality remains modest. Most of the dispersion in lifetime hours is generated after entry.

The last column of Table 5 uses the state vector five years after labor-market entry as the base period. The year-5 base period corresponds to age 23 for the median high school graduate in our sample, the same chronological cutoff used by Huggett et al. (2011). The importance of initial conditions rises sharply. With the full state vector, the explained share of lifetime earnings inequality nearly doubles, from 26.4 to 45.6 percent. For lifetime wealth, the share rises from 33.8 to 47.6 percent; and for lifetime hours, from 17.7 to 22.6 percent. Measuring "initial" conditions five years after entry, therefore, reassigns a substantial part of post-entry variation to initial heterogeneity, especially for earnings and wealth.[21]

State variables observed several years after entry are not purely initial endow-

[21] Our year-5 base-period estimate is lower than the roughly 60 percent reported by Huggett et al. (2011) for earnings. The two estimates are not directly comparable. Differences relative to their estimate may reflect model structure, data sources (PSID versus NLSY), sample design (their broader set of workers and cohorts versus our focus on high-school graduates from specific cohorts), and the timing of measurement.

ments. They already embed the effects of early-career shocks, labor-supply choices, human capital accumulation, and match sorting. Treating those states as initial conditions assigns part of the school-to-work transition to predetermined heterogeneity. Our framework allows us to separate the two, and the separation shows that a substantial share of what age-based decompositions attribute to initial heterogeneity is, in fact, produced during the early career. The next two subsections document the channels through which this happens.

5.2 Accounting for Wage and Earnings Growth and Dispersion

We next use the model to account for the sources of growth and dispersion in wages and earnings. The exercise is related to the decomposition in Menzio et al. (2016), who separate wage growth into the contributions of experience and match quality. We extend this logic in two directions. First, we apply the same accounting not only to mean wages but also to wage dispersion. Second, we study earnings, which adds hours worked as an additional source of both growth and inequality. The decomposition is a wage-equation accounting and therefore captures only the direct effect of each state on wages.[22]

The decomposition is implemented using annual averages of monthly log components among employed workers. We average monthly logs, rather than taking logs of annual averages, because the log-additive wage and earnings decompositions hold exactly at the monthly level in the simulated data.[23]

[22] As Subsection 5.3 will show, human capital also raises match quality indirectly, through the transition-rate and offer-quality channels – channels that this decomposition does not isolate. The human-capital and match-quality contributions reported below are therefore not fully independent: part of the match-quality contribution is itself produced by human capital operating through transitions and offers.

[23] Given the function form that we assumed for the wage equation, mean log wages decompose into the contributions of human capital, match quality, and the age component:

$$\Delta\mathbb{E}[\log w_t] = \Delta\mathbb{E}[\rho_{ws}\log s_t] + \Delta\mathbb{E}[\rho_{wm}\log m_t] + \Delta\alpha_t.$$

Wage dispersion decomposes into variance and covariance components:

$$\begin{aligned}\mathrm{Var}(\log w_t) &= \mathrm{Var}(\rho_{ws}\log s_t) + \mathrm{Var}(\rho_{wm}\log m_t)\\ &\quad + 2\mathrm{Cov}(\rho_{ws}\log s_t, \rho_{wm}\log m_t).\end{aligned}$$

The first term captures dispersion in human capital, the second captures dispersion in match quality, and the covariance term captures sorting between the two. A positive covariance indicates that workers with higher human capital hold better matches, amplifying wage inequality.

Earnings equal wages times hours, so log earnings decompose additively among employed workers:

$$\log e_{it} = \log w_{it} + \log h_{it},$$

Mean Growth Figure 6 reports the decomposition from labor-market entry. Mean wage growth reflects both skill accumulation and movement toward better matches. Over the first 20 years, mean log wages rise by 0.552 log points. The reported human-capital and match-quality components contribute 0.326 and 0.322 log points, respectively. These reported components omit the deterministic age component in the wage equation, which is negative over this horizon. Wage growth, therefore, reflects both skill accumulation and movement toward better matches, partly offset by the omitted age component. Mean earnings grow more strongly, by 0.777 log points, because hours also rise. Human capital and match quality contribute 0.326 and 0.322 log points, respectively, while hours contribute 0.225 log points. The deterministic age component contributes -0.096 log points.

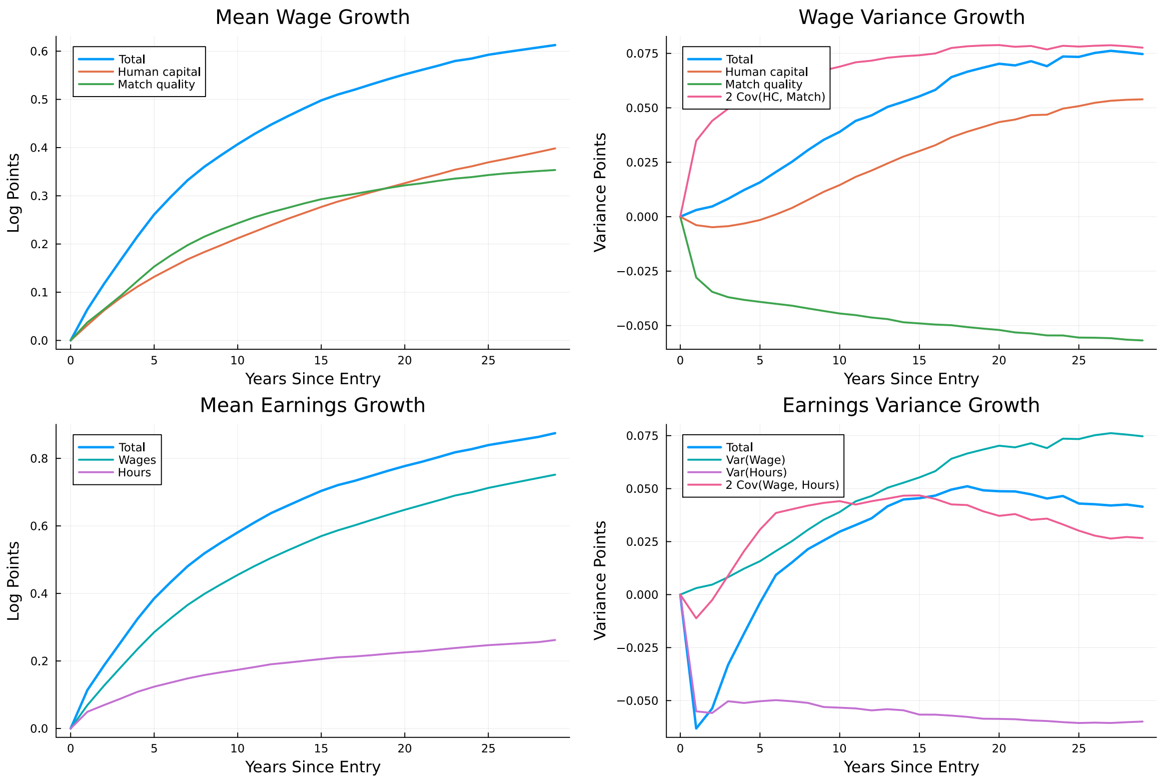


Figure 6: Wage and earnings growth and dispersion decompositions from labor-market entry.

Dispersion Growth Wage dispersion follows a different logic from mean growth. Variance rises from 0.108 at entry to 0.178 after 20 years, but the increase is not

and earnings dispersion satisfies

$$\text{Var}(\log e_t) = \text{Var}(\log w_t) + \text{Var}(\log h_t) + 2\text{Cov}(\log w_t, \log h_t).$$

This separates earnings dispersion into wage dispersion, hours dispersion, and the covariance between wages and hours.

driven by widening match-quality dispersion, which actually falls over the career. Instead, most of the increase is accounted for by the rising covariance contribution between human capital and match quality in the wage-variance decomposition. The underlying correlation between $\log s_t$ and $\log m_t$ is -0.46 at entry and becomes positive after five years. In words, workers with higher human capital increasingly hold better matches. Wage inequality grows because skills and jobs become aligned, not because either distribution widens in isolation.[24]

The sorting result reflects the channels documented in Subsection 5.3. Higher-human-capital workers face better offer distributions and lower separation risk, so skill accumulation translates endogenously into better matches. The positive covariance between s and m in the wage equation is the consequence of these channels operating on the joint state distribution. Without the transition-rate and offer-quality human-capital channels, the covariance would not rise.

Earnings dispersion is determined by wage and hours dispersion. Wage dispersion rises from 37 percent of earnings variance at entry to 53 percent after 20 years, while hours dispersion falls from 44 percent to 21 percent. The wage-hours covariance is positive and rising, so high-wage workers also work longer hours, but the decline in hours dispersion dampens the increase in earnings inequality. The hours dispersion falls because the search-driven component declines, but it stabilizes at a level determined by preference heterogeneity. Without preferences, it would fall further. Earnings variance, therefore, rises less sharply than wage variance.

Timing of Measurement We now show how the timing of measurement shapes the interpretation of these decompositions. Table 6 compares the accounting when the base period is labor-market entry with the accounting when the base period is five years after entry. To keep the comparison concise, the table reports changes over the next ten years from each base period.

The timing comparison reverses the interpretation of wage inequality growth. Measured from entry, the dominant force is sorting: the covariance between human capital and match quality contributes 176.8 percent of wage variance growth over the first ten years, while match-quality dispersion declines by 114.1 percent and

[24] The initial correlation between $\log s_0$ and $\log m_0$ is $\rho_{sm0} = -0.4592$ in the estimated model and is disciplined by observed dispersion in the initial workweek. Its contribution to log-wage variance is $2\rho_{ws}\rho_{wm}\mathrm{Cov}(\log s_t, \log m_t)$, which also depends on the wage loadings and the dispersions of human capital and match quality. The movement from negative to positive sorting occurs because higher-human-capital workers face better offer distributions and lower separation risk, causing the joint distribution of s and m to move toward positive covariance.

Table 6: Timing and the Accounting of Mean and Dispersion Growth

	From Entry		From Year 5	
Component	Contribution	Percent	Contribution	Percent
Panel A: Mean wage growth				
Human capital	0.212	52.1%	0.145	61.2%
Match quality	0.243	59.7%	0.140	59.0%
Total	0.407	100.0%	0.237	100.0%
Panel B: Wage variance growth				
Human capital	0.015	37.3%	0.032	80.2%
Match quality	-0.044	-114.1%	-0.010	-25.0%
Covariance	0.069	176.8%	0.018	44.8%
Total	0.039	100.0%	0.040	100.0%
Panel C: Earnings variance growth				
Wages	0.039	131.1%	0.040	80.1%
Hours	-0.053	-179.6%	-0.006	-12.8%
Covariance	0.044	148.4%	0.016	32.7%
Total	0.030	100.0%	0.049	100.0%

Note: The table compares ten-year changes using two different base periods. The entry-based columns measure changes from labor-market entry to year 10. The year-5 columns measure changes from year 5 to year 15 after entry. Panel A reports changes in mean log wages; Panels B and C report changes in variances. Percent contributions in Panel A are relative to total mean wage growth including the deterministic age component, which is not reported separately.

human-capital dispersion contributes only 37.3 percent. Measured from year 5, the human-capital component contributes 80.2 percent, and the covariance contribution falls to 44.8 percent. Starting later, therefore, makes wage inequality growth look like a story of widening human-capital dispersion, while starting from entry reveals it as a sorting story driven by the channels through which human capital affects labor-market position.

The same pattern appears for earnings, moderated by hours. Wage dispersion accounts for 131.1 percent of earnings variance growth from entry but only 80.1 percent from year 5, while the offsetting role of hours dispersion shrinks from -179.6 to -12.8 percent.

The implication for the broader inequality literature is the same as in Subsection 5.1. Decompositions that begin five years after entry treat the sorting between human capital and match quality, the convergence of employment states, and the early-career wage growth as predetermined, and miss the channels through which they were produced. Measuring workers from labor-market entry reveals that early-career sorting is itself a central source of wage inequality growth. We show in the next subsection that the sorting is generated by the transition-rate and offer-quality channels through which human capital shapes labor-market position. This is one of the central reasons to track workers from entry rather than from a fixed age cutoff: the early career is when these channels do most of their work, and a decomposition that starts later conditions on outcomes the framework is trying to explain.

5.3 Mechanisms and Counterfactual Economies

The previous sections used the estimated model to decompose life-cycle profiles and lifetime inequality. We now use counterfactual economies to isolate the mechanisms behind those results, identify the forces that drive early-career labor supply, and show why distinguishing among these mechanisms matters. We compare the baseline economy to three counterfactuals, each corresponding to a canonical specification in the prior literature.

The first counterfactual is a standard life-cycle search economy with initial productivity and preference heterogeneity but no human-capital accumulation. This is the limiting case of frictional search models that abstract from endogenous skill. The second is a standard human-capital economy in which all workers start employed, the initial match is permanent, and human capital evolves through labor supply and affects wages alone. This is the limiting case of life-cycle labor-supply models with human-capital accumulation but no search frictions, in the tradition of Heckman (1976), Imai and Keane (2004), and Keane and Rogerson (2015).[25] The third counterfactual shuts down preference heterogeneity to isolate its role in life-cycle hours dispersion, an issue emphasized by Bick et al. (2024).[26]

We further decompose the human-capital mechanism into three channels:

[25]Appendix C considers an additional portable-human-capital economy, in which accumulated human capital is no longer partly match-specific, and shows that allowing skills to travel across jobs strengthens rather than weakens the interaction between human capital and search.

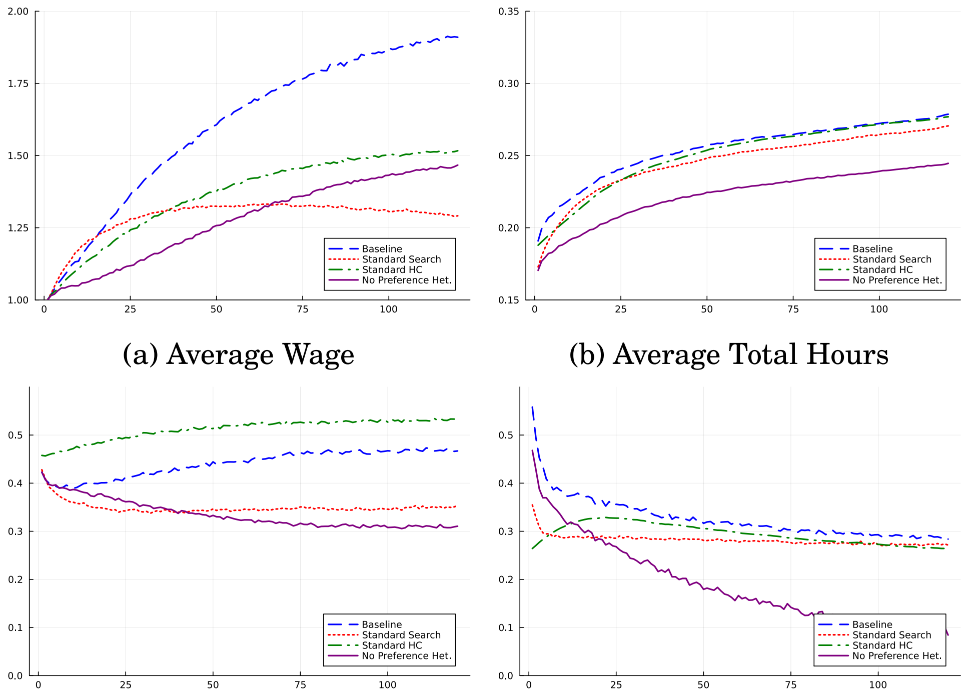
[26]In the standard-search economy, we set the human-capital accumulation, depreciation, and loss parameters to zero, and remove human capital from job-finding rates, separation rates, and offer-quality distributions. Wages still depend on the initial level of human capital. The offer-quality intercepts and residual variances are adjusted to preserve the unconditional initial mean and variance of offer quality implied by the baseline model. The constant monthly job-finding and separation rates are jointly chosen so that the economy converges to an employment rate of 0.95 and, starting from an initial employment rate of 0.75, generates an average employment rate of 0.80 during the first quarter. In the standard human-capital economy, all workers begin employed, on-the-job offers and separations are shut down, and the initial match is permanent. The initial match-quality distribution is renormalized to preserve initial average wages, so that the difference between this economy and the baseline reflects the absence of search rather than a change in entry-period wage levels. In the no-preference-heterogeneity economy, the initial and stationary dispersion in preferences is set to zero.

wages, transition rates, and the mean of the offer distribution. As in standard life-cycle labor-supply models with human-capital accumulation, labor supply has an intertemporal return: work today raises future human capital, so the return to working an additional hour is not only the current piece rate but also the expected future gains generated by skill accumulation (Heckman, 1976; Imai and Keane, 2004; Keane and Rogerson, 2015). In our framework, these future gains arise through three channels. The standard channel emphasized by the prior literature is that human capital raises future wages. The two additional channels are that human capital raises job-finding rates and lowers separation rates, and that it shifts the mean of the offer distribution upward, so that workers with more human capital draw from better job-quality distributions. We study each channel in isolation.[27]

Mechanism Comparison We start with Figure 7, which illustrates how the model's identification works. The top-left panel reports mean wages. All economies generate rising wages, with the baseline producing the fastest and highest growth because it combines all mechanisms. The bottom-left panel reports wage dispersion and shows a contrast. Economies with human capital exhibit increasing dispersion, whereas standard-search and no-preference-heterogeneity economies exhibit decreasing or flat dispersion. The standard-search economy converges to a stationary distribution of match quality because the high finding rate and low separation rate imply a flat dispersion for mid-career workers. Human capital is therefore essential for generating the observed increase in wage dispersion.

The top-right panel reports mean total hours – the product of the workweek and the employment rate. As with wages, all economies produce rising hours, but the path differs. The standard-search economy generates a sharp early-career increase from both margins: workers transition out of unemployment into employment, and rising match quality raises wages, which raises hours conditional on employment through the substitution effect. The increase flattens once the match-

[27] Each channel counterfactual starts from the baseline-calibrated parameters and shuts down a single channel. In the no-wage channel counterfactual, the wage elasticity with respect to human capital, ρ_{ws}, is set to zero, and the wage constant is renormalized to preserve the initial average wage. In the no-transition-rate channel counterfactual, the human-capital elasticities of the job-offer and separation rates, $\rho_{\lambda s}$ and $\rho_{\pi s}$, are set to zero, and their intercepts are adjusted to preserve average initial finding, job-to-job, and separation probabilities. In the no-offer-quality channel counterfactual, human capital is removed from the mean of the match-quality offer distributions, and the intercepts are adjusted to preserve average initial offer quality. The full no-human-capital-dynamics counterfactual shuts down accumulation, depreciation, and human-capital losses, while leaving the initial cross-sectional distribution of human capital and its static effects on wages, transition rates, and offer quality intact.



(a) Average Wage (b) Average Total Hours

(c) Wage Dispersion (d) Total Hours Dispersion

Figure 7: Counterfactual Economies

quality distribution stabilizes, since hours track wages through the substitution channel. The human-capital economy generates a smoother increase, since all workers are already employed and hours rise only through the workweek. The baseline lies between the two, reflecting the contributions of both mechanisms. The no-preference-heterogeneity economy generates a similar path to the baseline but at a lower level, because removing the dispersion in preference types shifts the average disutility of work.

The bottom-right panel reports total hours dispersion. Search frictions are essential for generating the high early-career hours dispersion observed in the data, echoing the role of the wedges identified by Kaplan (2012). In the standard-search economy, hours dispersion mirrors wage dispersion through the same substitution channel: as the match-quality distribution converges, so does the hours distribution. The human-capital economy, by contrast, generates rising hours dispersion early in the career – the opposite of the data – because rising human capital raises wages, which induces a substitution toward more hours, and dispersion in human-capital paths translates into dispersion in hours. Preference heterogeneity is essential for generating persistent hours dispersion through the life cycle, echoing Bick et al. (2024): without it, hours dispersion falls substantially as workers' employment and human-capital states converge.

We replicate the hours decomposition from Subsection 4.5 to understand how each mechanism contributes to the intensive and extensive margins of labor supply. Results are in Table 7 for 10 years after labor-market entry. In the baseline, hours growth is almost evenly split between the two margins: 56.5 percent comes from higher hours conditional on employment, and 43.1 percent comes from higher employment. The two canonical economies differ on this split.

In the standard-search economy, the intensive margin accounts for 94 percent of hours growth – a standard substitution effect, as rising match quality raises the return to working an additional hour. The extensive-margin contribution is 21 percent, but it is partly offset by a negative composition term of -15 percent, so their net contribution is small. The negative composition term arises because workers in the decomposition window have, on average, longer tenure and therefore higher hours, while newly hired workers have lower hours on average. The no-preference-heterogeneity economy is close to the baseline, with a slight shift toward the intensive margin, and the standard-human-capital economy has only the intensive margin active by construction.

Table 7: Labor Supply Decomposition at Year 10

Economy	Intensive	Extensive	Composition	Quadratic
Baseline	56.5%	43.1%	0.2%	0.3%
Standard Search	93.9%	20.7%	-15.4%	0.8%
Standard HC	100.0%	0.0%	0.0%	0.0%
No Pref. Het.	63.8%	38.4%	-3.9%	1.7%

Both canonical economies depart from the empirical split in distinct ways. The standard-search economy generates an extensive-margin contribution that is too small once composition is netted out. The standard-human-capital economy excludes the extensive margin entirely. The baseline matches the data by combining both: search frictions generate employment growth, and human-capital accumulation generates workweek growth.

Another way of examining labor-supply incentives is to compute, for each employed worker, the ratio of the marginal rate of substitution between leisure and consumption to the current wage,

$$\frac{\text{MRS}_{it}}{w_{it}} = \frac{\xi_{it} h_{it}^{1/\gamma} c_{it}}{w(s_{it}, m_{it})}. \tag{5}$$

A static labor-supply model predicts that this ratio equals one, since workers choose hours where the marginal cost of work (MRS) equals the marginal benefit (the wage). The ratio exceeds one when workers supply hours at which the static cost of work exceeds the static benefit. The wedge summarizes the strength of dynamic incentives – investment in future productivity, future labor-market prospects, and current resources when borrowing constraints limit consumption smoothing – that lead workers to choose hours beyond the static optimum. The presence of such a wedge is emphasized in Keane and Rogerson (2015), who argue that human-capital accumulation reconciles small static Frisch elasticities with the larger dynamic responsiveness implied by macro labor-supply behavior.

Imai and Keane (2004) plot this ratio to document the force of human-capital accumulation in their estimated model, reporting an MRS/wage ratio of 2.0 at age 20 that converges toward one over the life cycle. We do the same in Figure 8. The baseline and the human-capital economy both generate declining dynamic incentives over the life cycle, consistent with the pattern in Imai and Keane (2004). The standard-search economy generates a flat ratio close to one, indicating that an economy without human-capital accumulation produces essentially no dynamic wedge: workers supply hours at the static optimum. The declining profile in the economies with human capital reflects two forces. The marginal return to additional hours falls with age because (i) the human-capital production function has decreasing returns, and (ii) the remaining lifespan decreases with age. The no-preference-heterogeneity economy generates a wedge that closely tracks the baseline, indicating that preference heterogeneity has little effect on the average dynamic incentive.

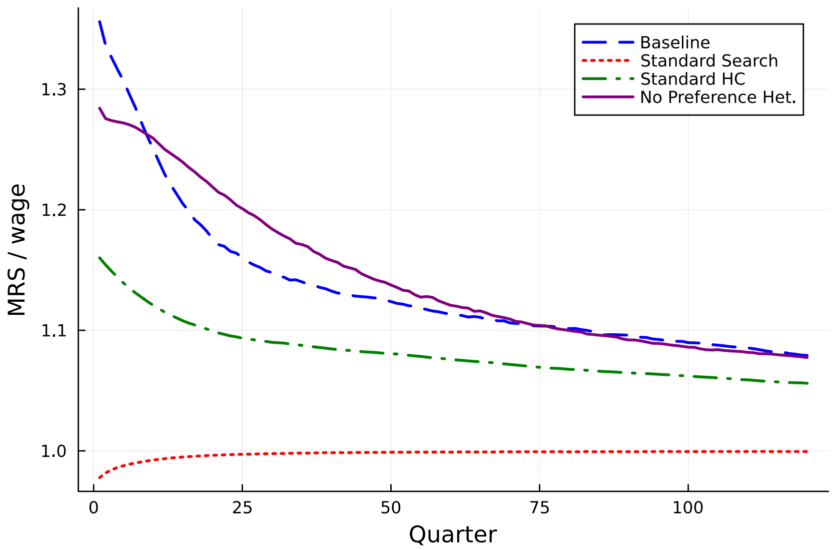


Figure 8: Marginal Rate of Substitution Relative to Wages

Table 8 quantifies the magnitudes. In the baseline, the average MRS/wage ratio is 1.253, 1.201, and 1.158 over the first five, ten, and twenty years. The remaining columns report the size of the wedge in each counterfactual relative to the baseline.

Table 8: Marginal Rate of Substitution: Baseline and Counterfactuals

		Dynamic Wedge Relative to Baseline		
Years	Baseline MRS/w	Standard Search	Standard HC	No Pref. Het.
5	1.253	-3.56%	48.76%	100.47%
10	1.201	-2.82%	53.16%	109.12%
20	1.158	-2.11%	57.85%	108.48%

Note: The first column reports the average MRS/wage ratio in the baseline economy over the horizon indicated by each row. The remaining columns report, for each counterfactual economy, the signed dynamic wedge relative to the signed baseline wedge, computed as $100 \times [((\text{MRS}/w)_{cf} - 1)/((\text{MRS}/w)_{base} - 1)]$. A value of 100 indicates a wedge equal to the baseline wedge, while a value close to zero indicates essentially no dynamic wedge. Negative values indicate that the average MRS/wage ratio in the counterfactual economy is below one.

Three results emerge. First, the standard-search economy generates essentially no dynamic wedge, and the MRS/wage ratio is essentially at one. Without human-capital accumulation, there is no return to working beyond the static optimum. Second, the standard-human-capital economy produces about half the baseline wedge – 49 percent in the first five years, rising to 58 percent at twenty years. The standard human-capital economy misses half the baseline wedge despite having full human-capital dynamics. The difference comes from the channels through which human capital affects labor-market outcomes beyond the wage. Third, the no-preference-heterogeneity economy produces a wedge essentially equal to the baseline, slightly larger at longer horizons.

Human Capital Channel Decomposition The preceding counterfactuals show that human capital is central for generating dynamic labor-supply incentives. We now decompose the human-capital mechanism into three channels: wages, transition rates, and the mean of the offer distribution. We do it by removing each channel separately, alongside a counterfactual that removes human-capital dynamics entirely while preserving initial human-capital heterogeneity. Figure 9 reports the MRS/wage ratio in each economy.

No single channel accounts for the baseline dynamic wedge. Removing human capital from wages substantially lowers the MRS/wage ratio, confirming that the standard wage channel is an important source of dynamic incentives. Remov-

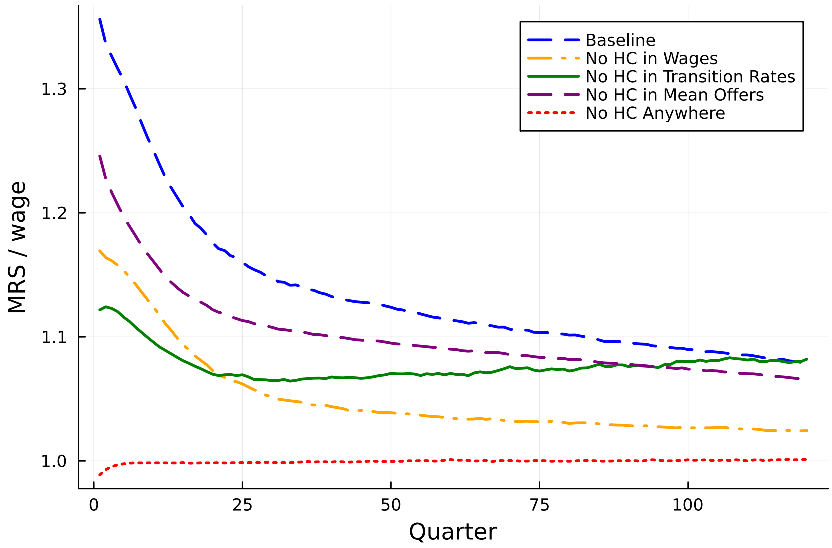


Figure 9: Marginal Rate of Substitution Relative to Wages

ing human capital from transition rates also lowers the wedge, especially early in the life cycle: higher human capital raises future job-finding rates and lowers separation risk, providing an additional motive for current labor supply. Removing human capital from the mean of the offer distribution has a smaller but non-trivial effect on the average wedge.

Table 9 quantifies the magnitudes. Under the same convention as Table 8, each column reports the wedge in the counterfactual as a percentage of the baseline wedge. Removing human capital from wages leaves 48 percent of the baseline wedge intact in the first five years and 39 percent at twenty years – equivalently, the wage channel accounts for roughly half the baseline wedge early in the career, with its share rising to nearly two-thirds at longer horizons. Removing human capital from transition rates leaves 38 percent of the wedge in the first five years, rising to 48 percent at twenty years. The transition-rate channel, therefore, accounts for slightly more than half the baseline wedge in the early career, with its share declining over time. Removing human capital from the mean of the offer distribution leaves 66 to 72 percent of the wedge, so the offer-quality channel accounts for the remaining 28 to 34 percent. Finally, the no-HC-dynamics economy produces an MRS/wage ratio essentially at one: without any human-capital accumulation, workers have no reason to supply hours beyond the static optimum.

Three patterns emerge from the channel decomposition. First, the wage channel and the transition-rate channel are each large and comparable in magnitude – both account for roughly half the baseline wedge in the early career, with their relative importance reversing over the life cycle. The two channels cross over in

Table 9: Human Capital Channel Decomposition

		Dynamic Wedge Relative to Baseline			
Years	Baseline	No HC in Wages	No HC in Trans. Rates	No HC in Mean Offers	No HC Anywhere
5	1.253	47.93%	38.18%	65.90%	-1.05%
10	1.201	43.47%	40.62%	68.41%	-0.96%
20	1.158	38.91%	48.29%	72.13%	-0.64%

Note: The first column reports the average MRS/wage ratio in the baseline economy over the horizon indicated by each row. The remaining columns report, for each counterfactual economy, the dynamic wedge as a percentage of the baseline wedge, computed as $100 \times [((\mathrm{MRS}/w)_{cf} - 1)/((\mathrm{MRS}/w)_{base} - 1)]$.

importance around the middle of the career: in the first decade the transition-rate channel dominates, and in the second decade the wage channel does. Second, the offer-quality channel is smaller but still meaningful: it accounts for about one-third of the dynamic wedge. Third, the channels are not additive substitutes. Each is necessary for the full baseline wedge, and the framework's contribution comes from having all three operate simultaneously.

The hours decomposition in Table 10 shows how the channels map into intensive and extensive margins. Removing human capital from wages leaves the decomposition close to the baseline (59.6 / 41.3 vs. 56.5 / 43.1): both margins remain active because human capital still affects transitions and offers. Removing human capital from transition rates collapses the extensive margin from 43.1 to 9.2 percent and pushes the intensive margin to 91.1 percent. In this economy, human capital still raises wages and improves the quality of future offers, but it no longer improves employment stability or job-arrival prospects. Thus, the transition-probability component of the incentive to work for better future jobs largely disappears, and hours growth becomes almost entirely intensive-margin. Removing human capital from mean offers leaves the intensive-extensive split close to the baseline (61.5 / 41.7), indicating that offer-quality effects shape the level of the dynamic wedge but not the margin on which it operates.

Our channel decomposition quantifies the importance of each mechanism and therefore helps identify which margins to consider when evaluating labor-supply responses to a given intervention. The magnitude of the aggregate dynamic wedge alone does not reveal the channels through which the return to work arises. The standard channel through which human capital raises wages means that policies affecting current and future pay alter dynamic labor-supply incentives. In addi-

Table 10: Labor Supply Decomposition at Year 10

Economy	Intensive	Extensive	Composition	Quadratic
Baseline	56.5%	43.1%	0.2%	0.3%
No HC in Wages	59.6%	41.3%	-1.1%	0.2%
No HC in Trans. Rates	91.1%	9.2%	-6.2%	5.9%
No HC in Mean Offers	61.5%	41.7%	-3.3%	0.1%
No HC Anywhere	84.0%	6.0%	6.8%	3.2%

tion, our model allows human capital to improve employment stability, job-finding prospects, and the quality of future offers. These additional channels mean that policies affecting hiring and separation frictions, search efficiency, or the distribution of job opportunities also alter labor-supply incentives. Understanding the sources of the dynamic wedge is especially important during the early career, when the wedge is largest.

6 Conclusion

This paper studies the early career as a period in which lifetime labor-market inequality is formed. We estimate a life-cycle model with an endogenous workweek, human-capital accumulation, match quality, and frictional labor markets, using profiles indexed by quarters since labor-market entry. The framework allows hours worked to shape human capital, and human capital to shape wages, job-finding rates, separation rates, and the quality of outside offers. Estimating the model from labor-market entry allows us to quantify the importance of the school-to-work transition for the measurement of lifetime inequality.

The estimated model yields three findings. First, the timing of measurement is decisive for assessing the role of initial conditions. Initial conditions explain 26 percent of lifetime earnings dispersion when measured at labor-market entry, but 46 percent when measured five years after entry. This 20-percentage-point gap shows that decompositions that begin after labor-market entry partly reclassify inequality produced during the school-to-work transition as predetermined heterogeneity. Second, wage inequality rises over the early career because human capital and match quality become increasingly aligned: the covariance between the two moves from negative at entry to positive after five years and accounts for most of the increase in wage variance. Third, human-capital accumulation is the dominant source of early-career labor-supply incentives, but only about half of these incentives operate through wages. The remaining incentives operate through future labor-market prospects, as higher human capital raises job-finding rates, lowers separation risk, and shifts workers toward better offer distributions. Together, the findings show that modeling the early career is essential for understanding lifetime labor-market outcomes.

Two extensions seem especially relevant. First, our model abstracts from heterogeneity in education and family structure. Incorporating these dimensions would allow the framework to speak more directly to differences across demographic groups, as in Pedtke (2026), who studies the intergenerational implications of family structure. Second, we model hours as a continuous choice, while in practice workers often face fixed-hour contracts, part-time arrangements, and other institutional constraints on adjustment along the intensive margin. Assessing the implications of these constraints would test the robustness of our dynamic-hours incentive conclusion, particularly given that Gimpelson (2024) shows that hours constraints account for a substantial share of the variation in lifetime hours in Germany.

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A Additional Empirical Results

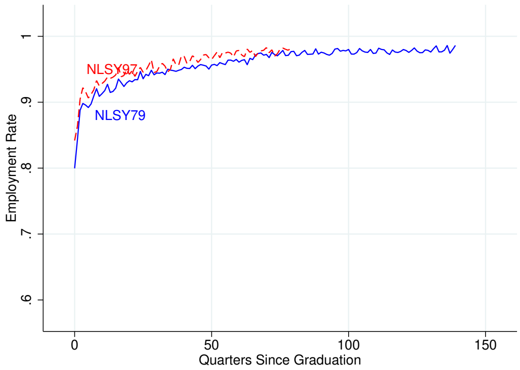
A.1 Sample Description

Table A1: Descriptive Statistics

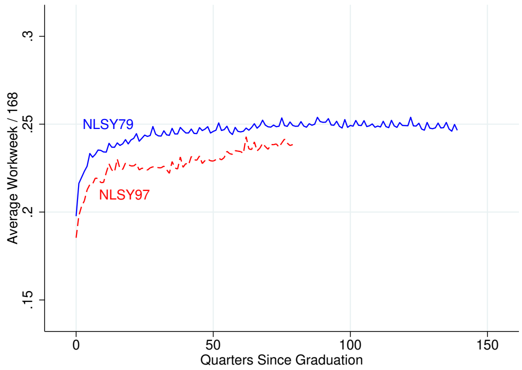
	Full Cohort	HS Graduates	Final Sample
Age (quarters)	133.92	137.94	147.35
	(50.29)	(51.24)	(52.09)
Sex	1.51	1.46	1.47
	(0.50)	(0.50)	(0.50)
Highest grade ever	14.01	11.96	11.99
	(3.15)	(0.19)	(0.09)
Total hours worked	377.29	361.03	373.73
	(1351.91)	(1170.26)	(1228.60)
Real hourly wage	12.56	10.74	11.02
	(8.44)	(6.36)	(6.46)
Fraction unemployed	0.05	0.06	0.06
	(0.19)	(0.21)	(0.21)
Fraction employed	0.70	0.66	0.67
	(0.44)	(0.45)	(0.45)

A.2 Comparison Between the Two Cohorts

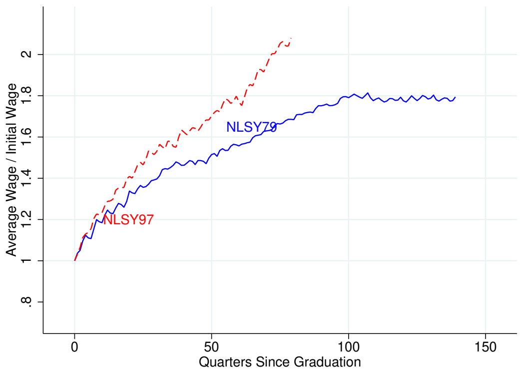
Figure A1 compares the mean profiles for the NLSY79 and NLSY97 cohorts. The main life-cycle patterns are similar across cohorts. Average wages are higher for the NLSY97 cohort than for the NLSY79 cohort, while the average workweek is lower for the NLSY97 cohort. By contrast, employment rates and the cumulative number of jobs held are very similar across cohorts.



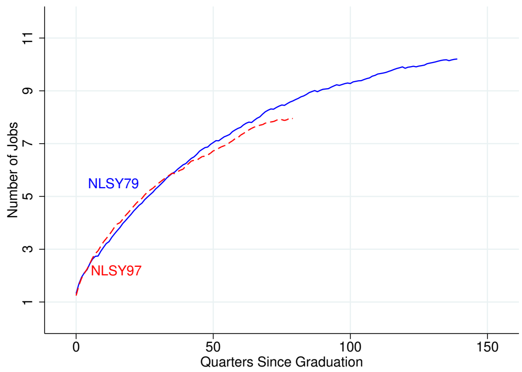
(a) Employment Rate



(b) Average Workweek



(c) Average Wage

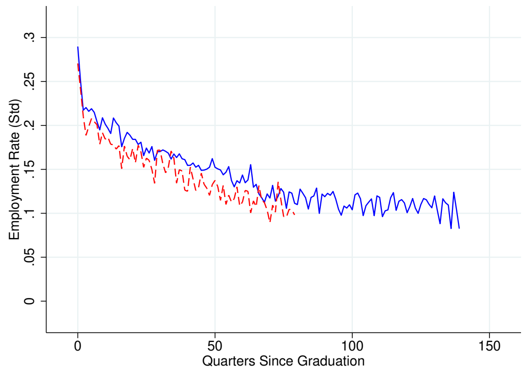


(d) Number of Jobs

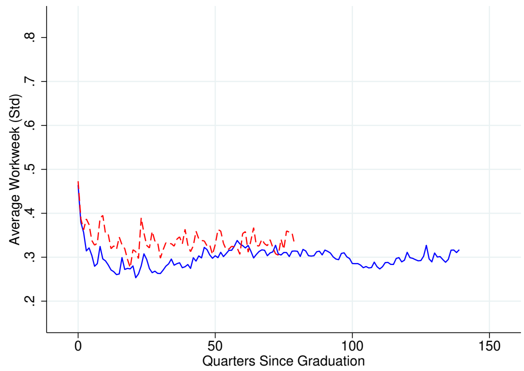
Figure A1: Mean Profiles by Cohort

Note: The figure compares mean profiles for the NLSY79 and NLSY97 cohorts. Employment is measured as the fraction of the quarter worked. The average workweek is normalized by 168 hours. Wages are normalized by the first observed wage in the profile. The number of jobs is measured as the cumulative number of distinct jobs held by the end of each quarter.

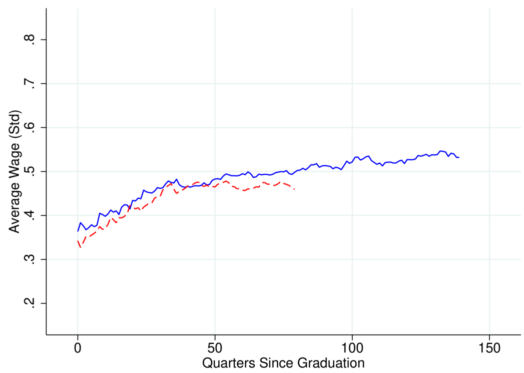
Figure A2 compares dispersion profiles across cohorts. The dispersion patterns are also highly similar. The standard deviations of the employment rate, average workweek, average wage, and cumulative number of jobs all track each other closely across the NLSY79 and NLSY97 cohorts. Thus, although the two cohorts differ in the levels of wages and workweeks, they display similar cross-sectional dispersion in the restricted sample.



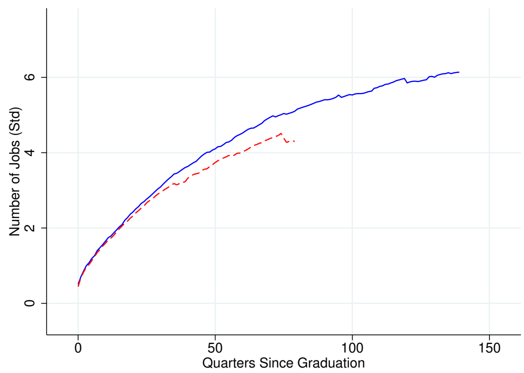
(a) Employment Rate (Std)



(b) Average Workweek (Std)



(c) Average Wage (Std)

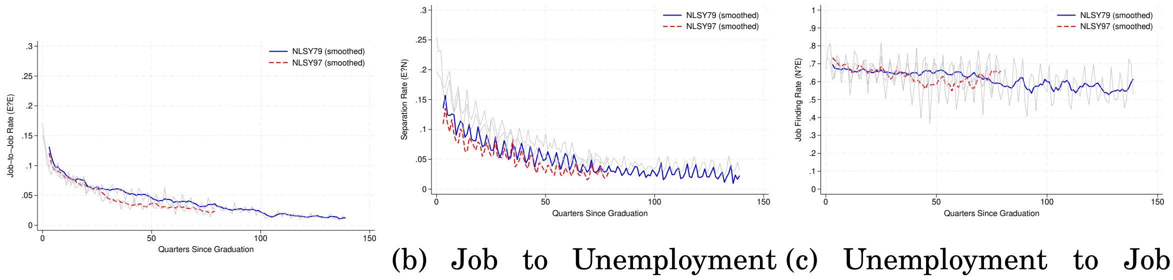


(d) Number of Jobs (Std)

Figure A2: Dispersion Profiles by Cohort

Note: The figure compares dispersion profiles for the NLSY79 and NLSY97 cohorts. The workweek and wage dispersion profiles are standard deviations of logs. Employment-rate dispersion is the standard deviation of the fraction of the quarter worked. Job dispersion is the standard deviation of cumulative jobs held.

Figure A3 shows that labor-market transition rates are also similar across cohorts. Job-to-job transitions, job-to-unemployment transitions, and unemployment-to-job transitions follow comparable life-cycle profiles in the NLSY79 and NLSY97. This reinforces the view that the main labor-market dynamics are stable across cohorts.



(a) Job to Job Rate

(b) Job to Unemployment Rate

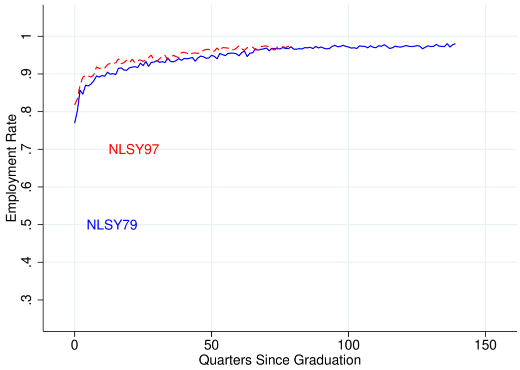
(c) Unemployment to Job Rate

Figure A3: Transition Rate Profiles by Cohort

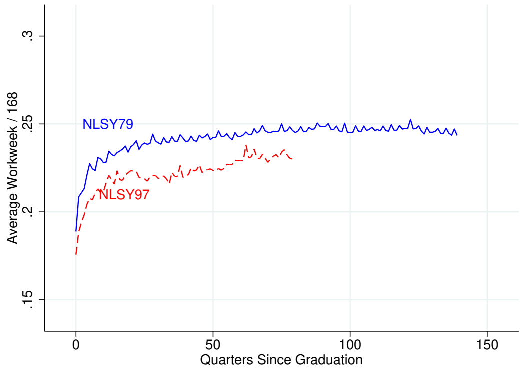
Note: The figure compares labor-market transition rates for the NLSY79 and NLSY97 cohorts. Transition rates are plotted by quarters since graduation.

A.3 Profiles Without Hours Restrictions

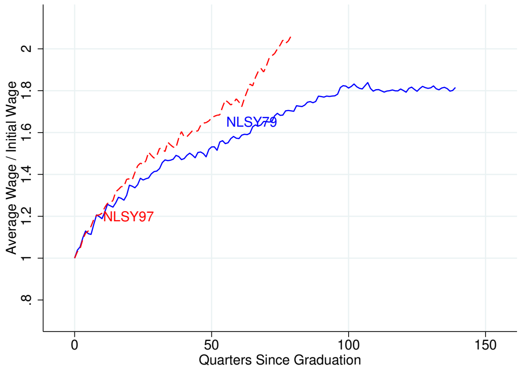
Figure A4 repeats the comparison without imposing the annual-hours restriction. The same broad differences across cohorts remain. Average wages are higher for the NLSY97 cohort, while the average workweek is lower. Employment rates and the cumulative number of jobs held remain similar across cohorts. Thus, removing the hours restriction does not materially change the comparison of mean profiles.



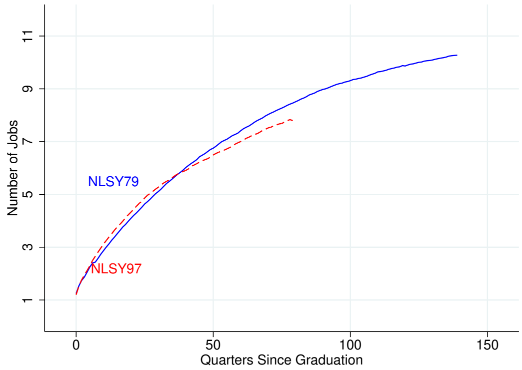
(a) Employment Rate



(b) Average Workweek



(c) Average Wage

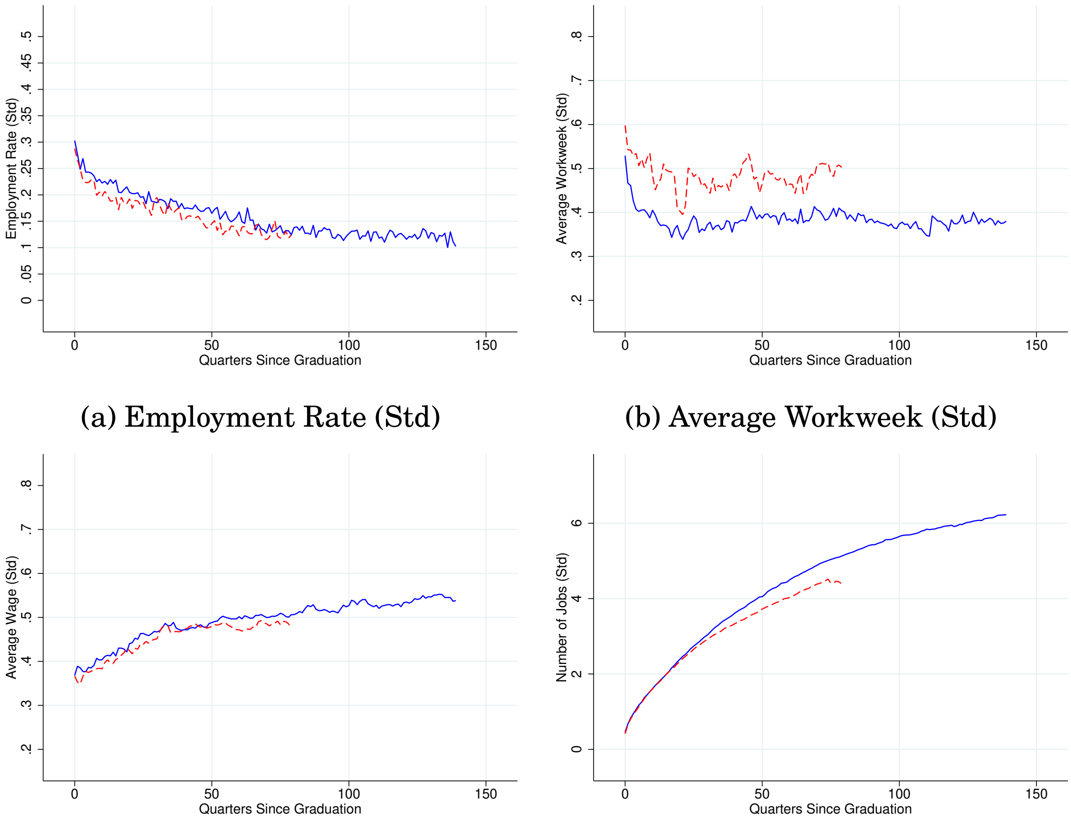


(d) Number of Jobs

Figure A4: Mean Profiles Without Hours Restrictions

Note: The figure plots mean profiles without restricting the sample to individuals with annual hours between 520 and 5,200. The NLSY79 is represented by solid lines and the NLSY97 by dashed lines.

Figure A5 shows a more interesting pattern for dispersion. Employment-rate dispersion, wage dispersion, and dispersion in cumulative jobs remain similar across cohorts. By contrast, workweek dispersion is higher for the NLSY97 cohort than for the NLSY79 cohort. This differs from Figure A2, where workweek dispersion is similar across cohorts in the restricted sample. The comparison suggests that the Kaplan (2012)-style pattern in workweek dispersion is stable across cohorts among attached workers, but changes once less attached workers are included. Hence, the cohort difference in unrestricted workweek dispersion appears to be driven by workers with weaker labor-market attachment.



(a) Employment Rate (Std)

(b) Average Workweek (Std)

(c) Average Wage (Std)

(d) Number of Jobs (Std)

Figure A5: Dispersion Profiles Without Hours Restrictions

Note: The figure plots dispersion profiles without restricting the sample to individuals with annual hours between 520 and 5,200. The workweek and wage dispersion profiles are standard deviations of logs.

B Derivation of the Hours Decomposition

This appendix derives the decomposition used in Section 4.5 to quantify the contributions of the intensive and extensive margins to the rise in hours worked after labor-market entry. The intensive margin is the average workweek. The extensive margin is the number of weeks worked. Because the panel is unbalanced, we also include a composition term that captures entry, exit, and changes in sample weights. Finally, because the decomposition is in discrete time, we keep a quadratic interaction term.

Idea: What would be the total hours worked if the intensive margin (or the extensive margin) is held constant? Perform a discrete-time approximation of the total derivative.

Notation:

- H_t = the average total hours in period t, where period t is measured in quarter since graduation. Its definition is: $H_t = \sum_{i \in E_t} \omega_{it} h_{it} n_{it}$.
- S_t = the set of individuals observed in period t.
- ω_{it} is the weight of worker i in period t. We use $\omega_{it} = 1/N_t$ in the paper.
- h_{it} = the average workweek in that period for worker i.
- n_{it} = the number of weeks worked in that period by worker i.

Implementation issues:

- Because the sample is not a balanced panel, the sample weights change across periods. Changes in the sample composition, including workers' entry, exit, and weight changes, need to be tracked.
- By performing a discrete approximation of the total derivative, second-order terms do not cancel out. They need to be tracked.
- Also, because of the approximation of the total derivative, changes need to be evaluated in a base period. The Laspeyres index is used, i.e., period $t-1$ is the base period.

Derivation Sketch:

The derivation proceeds by following these steps:

- First, decompose $H_t - H_{t-1}$ into the intensive and extensive margins.
- Second, sum $H_t - H_{t-1}$ for $t \in \{1, \ldots, T\}$ to find the contributions of the intensive and extensive margins over an extended period.

Formal Derivation:

A change in average total hours between periods t and $t-1$, $H_t - H_{t-1}$, can be decomposed as follows:

$$
\begin{aligned}
H_t - H_{t-1} &= \sum_{i \in E_t} \omega_{it} h_{it} n_{it} - \sum_{i \in E_{t-1}} \omega_{it-1} h_{it-1} n_{it-1} \\
&= \sum_{i \in E_t} \omega_{it} h_{it} n_{it} - \sum_{i \in E_{t-1}} \omega_{it-1} h_{it-1} n_{it-1} \pm \sum_{i \in E_{t\cap t-1}} \omega_{it} h_{it-1} n_{it-1} \\
&= \sum_{i \in E_{t\cap t-1}} \omega_{it} \left(h_{it} n_{it} - h_{it-1} n_{it-1} \right) \\
&\quad + \underbrace{\sum_{i \in E_{t\cap t-1}} \left(\omega_{it} - \omega_{it-1} \right) h_{it-1} n_{it-1} + \sum_{i \in E_t \setminus E_{t-1}} \omega_{it} h_{it} n_{it} - \sum_{i \in E_{t-1} \setminus E_t} \omega_{it-1} h_{it-1} n_{it-1}}_{\text{composition change}} .
\end{aligned}
$$

The last terms capture the impact of weight changes, as well as entry and exit. We bunch them together in a *composition change* term. The set $E_{t\cap t-1}$ is composed of all workers that are present in the sample in both periods t and $t-1$, or, more formally, $E_{t\cap t-1} = \{i \mid i \in E_t \cap E_{t-1}\}$.

Continuing with the derivation, $H_t - H_{t-1}$ can be further decomposed as:

$$
\begin{aligned}
H_t - H_{t-1} &= \sum_{i \in E_{t \cap t-1}} \omega_{it} \left(h_{it} n_{it} - h_{it-1} n_{it-1} \right) + \textit{comp. change} \\
&= \sum_{i \in E_{t \cap t-1}} \omega_{it} \left(h_{it} n_{it} - h_{it-1} n_{it-1} \right) \pm \sum_{i \in E_{t \cap t-1}} \omega_{it} h_{it} n_{it-1} + \textit{comp. change} \\
&= \sum_{i \in E_{t \cap t-1}} \omega_{it} \left(h_{it} - h_{it-1} \right) n_{it-1} + \sum_{i \in E_{t \cap t-1}} \omega_{it} h_{it} \left(n_{it} - n_{it-1} \right) + \textit{comp. change} \\
&= \sum_{i \in E_{t \cap t-1}} \omega_{it} \left(h_{it} - h_{it-1} \right) n_{it-1} + \sum_{i \in E_{t \cap t-1}} \omega_{it} h_{it} \left(n_{it} - n_{it-1} \right) \\
&\pm \sum_{i \in E_{t \cap t-1}} \omega_{it} h_{it-1} \left(n_{it} - n_{it-1} \right) + \textit{comp. change} \\
&= \underbrace{\sum_{i \in E_{t \cap t-1}} \omega_{it} \left(h_{it} - h_{it-1} \right) n_{it-1}}_{\text{intensive margin}} + \underbrace{\sum_{i \in E_{t \cap t-1}} \omega_{it} h_{it-1} \left(n_{it} - n_{it-1} \right)}_{\text{extensive margin}} \\
&+ \underbrace{\sum_{i \in E_{t \cap t-1}} \omega_{it} \left(h_{it} - h_{it-1} \right) \left(n_{it} - n_{it-1} \right)}_{\text{quadratic term}} + \textit{comp. change} \quad .
\end{aligned}
$$

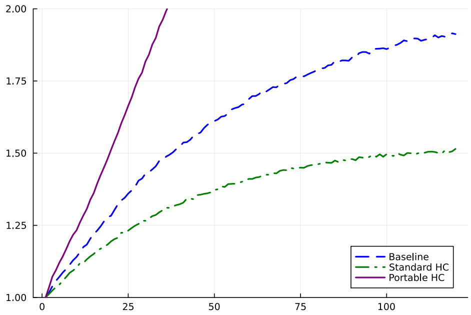
The first term refers to the intensive margin. It answers how much the increase in the workweek between t and $t-1$ increased total hours worked, holding the number of weeks worked in a year constant at its $t-1$ value. The second term refers to the extensive margin. Similarly, it answers how much the increase in weeks worked in a year between t and $t-1$ increased total hours worked, holding the workweek constant at its $t-1$ value. The quadratic term captures second-order changes. It is positive when weeks worked and the workweek tend to move together.

C The Role of Match-Specific Human Capital

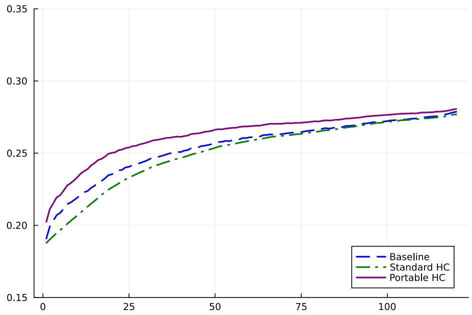
This appendix evaluates the role of match-specific human capital. The main text compares the baseline economy to canonical search and human-capital benchmarks. Here we ask a related question: if human capital is portable across jobs, does a model with human capital and search resemble the canonical standard human-capital model?

In the baseline model, human capital is partly match-specific. Workers lose a fraction of their accumulated skills when they separate or change jobs, as captured by the loss function $\phi(s) = Bs^{\nu}$. We construct a portable-human-capital counterfactual by shutting down these losses and removing match quality from the human-capital production function. Human capital remains productive in wages, job-finding rates, separation rates, and the mean of the offer distribution, but it is no longer tied to the current match. We compare this economy to the baseline and to the standard human-capital economy, in which all workers start employed, the initial match is permanent, and human capital affects wages alone.

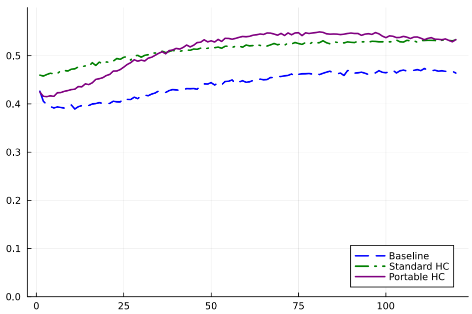
Figure C1 reports the life-cycle profiles. The portable-human-capital economy does not collapse to the canonical standard-human-capital benchmark. Instead, portability strengthens the interaction between human capital and search. Because workers can carry accumulated skills across jobs, current labor supply raises not only future wages but also future labor-market prospects without the offsetting risk of losing match-specific capital when moving. This raises the return to early-career work and changes both wage and hours dynamics.



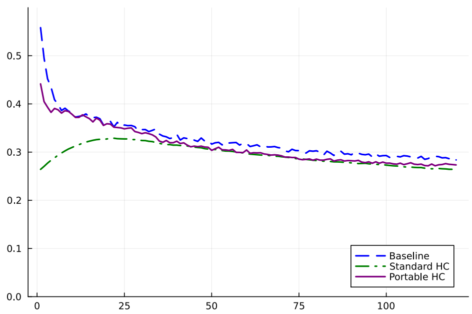
(a) Average Wage



(b) Total Hours



(c) Average Wage (Std)



(d) Total Hours (Std)

Figure C1: Human-Capital Portability

Figure C2 reports the marginal rate of substitution relative to wages. The portable-human-capital economy generates a substantially larger dynamic wedge than the baseline. This is the central result of the exercise. Portability makes human-capital investment more valuable because workers keep the returns to skill accumulation when they move up the job ladder. In this sense, search does not merely add an extensive margin to the standard human-capital model; it changes the return to human-capital investment itself.

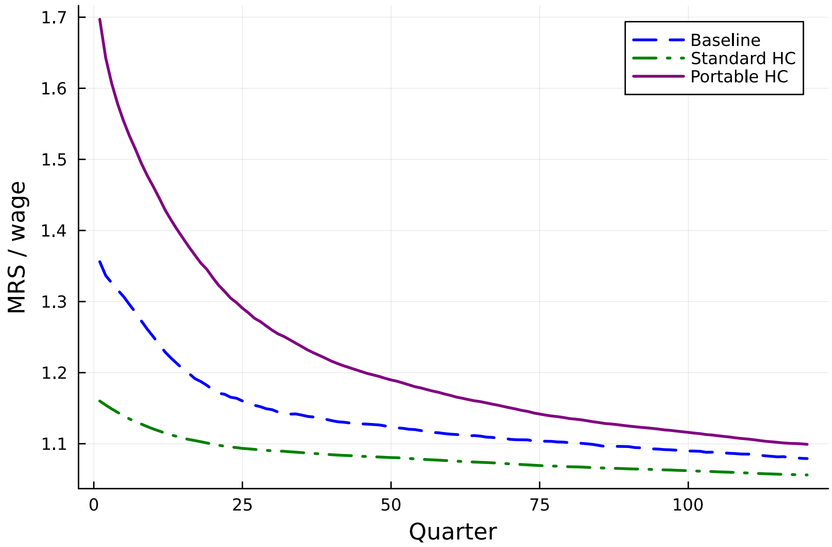


Figure C2: Marginal Rate of Substitution Relative to Wages

Table C1 quantifies the size of the dynamic wedge. The standard-human-capital economy produces about one-half of the baseline wedge, as in the main text. By contrast, the portable-human-capital economy produces a wedge that is 186 percent of the baseline over the first five years, 182 percent over the first ten years, and 170 percent over the first twenty years. The wedge is therefore not only larger than in the standard-human-capital benchmark, but also larger than in the baseline.

Table C1: Human Capital Portability

		Dynamic Wedge Relative to Baseline	
Years	Baseline MRS/w	Standard HC	Portable HC
5	1.253	48.76%	186.42%
10	1.201	53.16%	182.45%
20	1.158	57.85%	169.80%

The hours decomposition in Table C2 shows that portability also changes the margin through which labor supply grows. In the standard-human-capital economy, all growth comes from the intensive margin by construction. In the portable-human-capital economy, the split is almost even: 49.4 percent of hours growth comes from higher hours conditional on employment, and 50.1 percent comes from higher employment. Thus, portable human capital with search does not look like the canonical standard-human-capital model.

Table C2: Labor Supply Decomposition at Year 10: HC Portability

Economy	Intensive	Extensive	Composition	Quadratic
Baseline	56.50%	43.10%	0.15%	0.25%
Standard HC	100.00%	0.00%	0.00%	0.00%
Portable HC	49.41%	50.12%	-0.89%	1.37%

The conclusion is that match-specificity disciplines the dynamic incentive to work. When human capital is portable, workers retain a larger share of the return to skill accumulation when changing jobs. The result is an economy with a much larger dynamic wedge than the baseline and an extensive-margin contribution that remains central. Search with portable human capital, therefore, does not reduce to the canonical standard-human-capital model; it generates stronger incentives than either benchmark because human-capital accumulation and job search reinforce one another.